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CENTRAL UNIVERSITY OF TECHNOLOGY, FREE STATE

WORK-INTEGRATED LEARNING

Module Code: ITC327W

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Nama Nameng: A Mobile and Web-Based Ordering, Payment and Queue Management System for Nama Nameng

External Stakeholder: Nama Nameng (Chisa Nyama) — Owner: Doja

Digital Vanguard

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Table of Contents

- 1. Cover Page and Group Members 3
- 2. Project Title 3
- 3. Problem Statement 3
- 4. Purpose, Aim and Objectives 4
- 5. Project Scope 5
- 6. Deliverables and Acceptance Criteria 5
- 7. Exclusions, Constraints and Assumptions 6
- 8. Stakeholders and User Groups 6
- 9. Current Process and Diagram 7
- 10. Product Perspective 8
- 11. Main Flutter, ASP.NET and Supabase Functions 9
- 12. Functional Requirements 9
- 13. Non-Functional Requirements 10
- 14. System Requirements 11
- 15. Integration Requirements 11
- 16. Requirements Prioritisation 12
- Requirements Change Log 12
- Refined Project Scope 13
- Feasibility Study 13
- Project Risk Register 14
- SWOT Analysis 15
- 17. Appendices: Stakeholder Evidence (A–G) 15
  - Appendix A: Interview Notes 16
  - Appendix B: Requirements Confirmation Evidence 17
  - Appendix C: Site Visit / Photographic Evidence 18
  - Appendix D: Communication Log 19
  - Appendix E: Supabase Setup 20
  - Appendix F: Microsoft Project Plan 22
  - Appendix G: GitHub Repository Link 23
- 18. Phase 2: System Architecture and Design 24
  - 18.1 System Architecture Diagram 24
  - 18.2 Use Case Diagram 25

- 19. Sequence Diagrams 26
- 20. Entity Relationship Diagram (ERD) 31
- 21. Flutter Interface Designs 32
- 22. ASP.NET Interface Designs 33
- 24. Navigation Flow Diagram 34
- 25. Phase 2 Stakeholder Design Review 36
- 26. Microsoft Project 37
- 27. GitHub 38
- 28. Project Progress Tracker 39
- 29. Phase 3: Implementation and Final Prototype 40
  - 29.1 Overview of Implemented Components 40
  - 29.2 Shared Supabase Backend 40
  - 29.3 Flutter Mobile Application 41
  - 29.4 ASP.NET Web Application 41
  - 29.5 Requirements-to-Function Status 42
  - 29.6 Shared Processes Between Flutter and ASP.NET 43
  - 29.7 Screenshots of Implemented Interfaces and Shared Processes 44
  - 29.8 Approved Changes Made During Implementation 46
  - 29.9 Known Limitations and Incomplete Functions 48
  - 29.10 Security and Credential Protection 48
  - 29.11 Updated Risk Register 48
  - 29.12 Project Records, Source Code and Repository 49
  - 29.13 Work Allocation, Demonstration Plan and Remaining Work 50

1. Cover Page and Group Members

The cover page of this document identifies the institution, module, project title, external stakeholder and the group members responsible for this Work-Integrated Learning (WIL) project. Full names, student numbers and individual roles of all group members are recorded in the table on the cover page.

2. Project Title

Nama Nameng: A Mobile and Web-Based Ordering, Payment and Queue Management System for Nama Nameng

The project title reflects the core purpose of the system: connecting customers digitally to a chisa nyama business that currently relies entirely on in-person, walk-up ordering.

3. Problem Statement

3.1 Stakeholder and Organisation

Nama Nameng is a local chisa nyama that prepares and sells grilled meat and accompanying sides to walk-in customers in the surrounding community.

3.2 Current Situation

Nama Nameng currently operates using a fully manual, in-person process. Customers must physically travel to the premises, join a queue, and wait to reach the counter before they can even place an order. There is no app or web-based ordering option available; they only take orders using the owner's personal WhatsApp. The business has no website, no online menu, and no digital presence beyond informal word-of-mouth recommendations from existing customers.

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3.3 The Problem

- Customers experience long, visible physical queues, especially during peak trading times (weekends, lunch hours and paydays).
- There is no way for a customer to place an order in advance, browse the menu remotely, or check what is currently available before arriving.
- The business has no digital record of orders, sales trends, or customer preferences, making it difficult to plan stock or staffing.

  Marketing relies entirely on word-of-mouth, which limits the business's ability to reach new customers outside its immediate local network.
- Order-taking is manual and verbal, which increases the risk of miscommunication, forgotten items and incorrect orders.

3.4 Affected Users

- Walk-in and repeat customers of Nama Nameng, who lose time standing in queues.
- The owner/manager of Nama Nameng, who cannot track orders, sales or customer demand digitally.
  - Kitchen/braai staff, who receive orders verbally with no queue or ticketing system.
- Potential new customers in the surrounding area who are unaware of the business due to its reliance on word-of-mouth marketing.

3.5 Consequences and Need for Improvement

If the current manual process continues unchanged, Nama Nameng risks losing customers to competitors who offer more convenient ordering options, continues to miss opportunities to grow its customer base beyond its immediate community, and remains unable to make data-informed decisions about its menu, stock and staffing. A digital ordering and queue management solution is needed to reduce queuing time, extend the business's reach, and provide the owner with visibility over orders and sales.

4. Purpose, Aim and Objectives

4.1 Purpose

The purpose of this project is to design and develop a digital ordering and queue management solution for Nama Nameng, replacing the current fully manual, word-of-mouth-driven process with a mobile application for customers and a web-based management dashboard for the business, supported by a shared cloud backend.

4.2 Aim

To reduce customer queuing time and improve order accuracy at Nama Nameng by enabling customers to browse the menu and place orders digitally, while giving the business owner a simple, reliable way to manage orders, menu items and basic sales information online.

4.3 Objectives

- To develop a cross-platform mobile application (Flutter) and a web-based management dashboard (ASP.NET) that allows customers to view the menu, place and pay for orders, and track order status remotely, and allows the Owner/staff to manage the menu, view and update incoming orders, and view basic order/sales history.
- To implement a shared backend (Supabase) that stores menu, order and user data, and synchronises data in real time between the mobile app and the web dashboard.

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- To reduce physical queuing by allowing customers to place orders ahead of arrival and receive a notification or status update when their order is ready.

- To provide the business with a basic digital record of orders that can support future marketing and stock-planning decisions.

5. Project Scope

This initial SRS defines the scope of the Nama Nameng system as it will be developed for the WIL project prototype. The scope focuses on ordering, menu management and queue-visibility functionality that directly addresses the problems identified with Nama Nameng.

5.1 In Scope

- A cross-platform mobile application (Flutter) and a web-based management dashboard (ASP.NET) allowing customers to view the menu, place and pay for orders, and track order status remotely, and allowing Owner/staff to manage the menu, view and update incoming orders, and view basic order/sales history.
- A shared Supabase backend providing the database, authentication and real-time synchronisation between the mobile app and web dashboard.
- Basic digital queue/ticket visibility so customers know their order position or estimated wait.

  Basic order history and simple sales overview for the owner.

6. Deliverables and Acceptance Criteria

Deliverable	Acceptance Criteria	Owner
Flutter mobile app & ASP.NET web app	Allows customers to view the menu, place and pay for orders, and track order status remotely; allows Owner/staff to manage the menu, view/update incoming orders and view basic order/sales history.	Group
Supabase backend (database & auth)	All order, menu and user data is stored centrally, and changes made from either app sync correctly and promptly across both platforms.	Group
Deliverable	**Acceptance Criteria**	**Owner**

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Digital menu	Menu items, descriptions and prices are accurate, current, and reflect what Nama Nameng actually offers, as confirmed by the stakeholder.	Group
Order status / queue visibility	Customer can see their order's current status (e.g. Received / Preparing / Ready) after placing it.	Group
Initial and final SRS documentation	SRS accurately reflects stakeholder-confirmed requirements and is approved/acknowledged by the stakeholder.	Group

7. Exclusions, Constraints and Assumptions

7.1 Exclusions

- Integration with third-party delivery platforms (e.g. Uber Eats, Mr D Food) is excluded from this initial prototype.
- Integrated card/online payment gateway processing is excluded from the initial prototype; payment recording will be simulated or limited to cash-on-collection/manual confirmation unless the stakeholder confirms otherwise.
- Loyalty/rewards programmes, advanced analytics and multi-branch support are excluded from this version.
- Native iOS-specific and Android-specific features beyond what Flutter provides by default are excluded.

7.2 Constraints

- The project must be completed within the academic semester timeframe set by the ITC327W module.
- Development is limited to the technology stack specified by the module: Flutter (mobile), ASP.NET (web), and Supabase (backend/database).

  Access to the stakeholder for requirement confirmation is limited to arranged visits/meetings and is dependent on the stakeholder's business operating hours.

7.3 Assumptions

- Nama Nameng has, or is willing to obtain, basic internet connectivity (Wi-Fi/mobile data) at the premises to run the staff web dashboard.
- Customers placing orders through the mobile app have access to a smartphone and mobile data.
- The stakeholder will remain available for periodic feedback and requirement confirmation throughout the project.
- The menu and pricing information provided by the stakeholder during initial engagement is accurate and will be confirmed before final development.

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8. Stakeholders and User Groups

Stakeholder / User Group	Description	Interest / Involvement
Nama Nameng (Owner/Manager)	Primary external stakeholder; owns and operates the chisa nyama business.	Provides requirements, confirms scope, approves menu content and system behaviour.
Kitchen/Braai Staff	Prepares food and currently receives orders verbally.	Will use the order-status/queue features of the web & app dashboard.
Customers (Walk-in & Repeat)	Community members who purchase food from Nama Nameng.	End users of the Flutter mobile app & ASP.NET web app; primary beneficiaries of reduced queuing.
Stakeholder / User Group	**Description**	**Interest / Involvement**
Project Group (Students)	WIL project team responsible for design, development and delivery.	Gathers requirements, designs and builds the Flutter, ASP.NET and Supabase system.
ITC327W Module Facilitator	Assessor of the WIL project.	Reviews and assesses project deliverables, including this SRS.

Stakeholder needs

The following needs were identified directly through engagement with Doja (Nama Nameng owner) during the initial stakeholder meeting (see Appendix A) and confirmed as accurate for this project (see Appendix B). These needs directly inform the functional and non-functional requirements in Sections 12 and 13.

- Reduce the time customers spend standing in physical queues, particularly during peak periods (weekends, lunch hours, paydays).
- Provide a way for customers to see the menu and place an order before or without needing to be physically present at the counter.
- Reduce order-taking errors caused by verbal, in-person communication between customers, counter staff and the kitchen/braai area.
- Give the owner a simple way to manage the menu (add, edit, remove items and prices) without needing technical assistance.

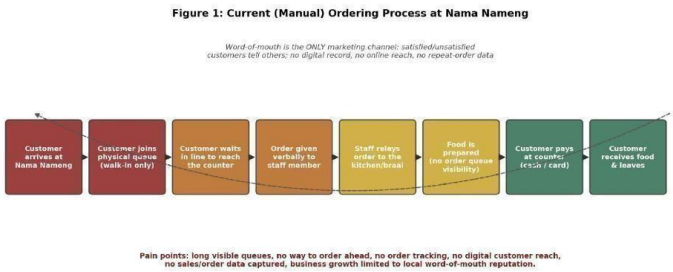
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- Give the owner visibility of incoming orders and the ability to update order status so customers know when their food is ready.
- Keep the solution simple and low-cost to operate, given the business's current reliance on informal, low-tech processes (WhatsApp, verbal ordering) and limited digital literacy among staff.
- Retain flexibility around payment — the business currently operates on a cash/in-person payment basis, and a full online payment gateway is not a priority need at this stage.
- Build a basic digital record of orders over time, to support future decisions on stock and staffing (a need the business cannot currently meet at all).

9. Current Process and Diagram

9.1 Current Process Description

Nama Nameng's current ordering process is entirely manual and in-person, as illustrated in Figure 1 below.

- A customer physically arrives at the premises; there is no remote ordering option. ● The customer joins a physical queue and waits their turn to reach the counter.
- Once at the counter, the customer verbally tells a staff member what they would like to order.
- The staff member relays the order to the kitchen/braai area verbally, with no ticket or digital queue reference.
- Food is prepared with no visibility to the customer of their position in the preparation queue.
  The customer pays at the counter (cash or card) and waits for the food to be ready.
- The customer receives their food and leaves; word-of-mouth sharing is currently the business's only form of marketing.



9.2 Key Pain Points Identified

- Long, visible physical queues, particularly at peak times.
- No way to check the menu remotely.
- No digital record of orders, sales or customer information.
- Marketing and customer growth limited entirely to word-of-mouth.

10. Product Perspective

Nama Nameng is a new, self-contained digital system developed specifically to replace Nama Nameng's manual ordering process. It is not a modification of an existing system.

The system consists of these integrated components:

- A Flutter-based mobile application & an ASP.NET web app used by customers to browse the menu, place orders and track order status. Staff use the aforementioned flutter and ASP.NET web-app to manage the menu and process incoming orders.
- A shared Supabase backend that provides the database, authentication and real-time data synchronisation connecting both front-ends.

The system will operate independently of any existing third-party ordering platforms and will be tailored specifically to Nama Nameng's menu, operating hours and day-to-day workflow, as confirmed through stakeholder engagement.

11. Main Flutter, ASP.NET and Supabase Functions

11.1 Flutter Mobile Application & ASP.NET Web (Customer-Facing)

- Customer registration and login.
- Browse digital menu with item descriptions, categories and prices.
- Add items to a cart and place an order.
- View real-time order status (e.g. Received, Preparing, Ready for Collection).

Page 9 of 24

Figure 1: Current (Manual) Ordering Process at Nama Nameng — reproduced from the Final Phase 1 SRS

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- Browse digital menu with item descriptions, categories and prices.
- Add items to a cart and place an order.
- View real-time order status (e.g. Received, Preparing, Ready for Collection).
- View basic order history for the logged-in customer.
- Receive in-app notifications/status updates on order progress.

11.2 Flutter Mobile Application & ASP.NET Web Dashboard (Owner/Staff-Facing)

- Staff/owner authentication and secure login.
- Add, edit, and remove menu items, categories and prices.
- View incoming orders in real time as they are placed.
- Update order status so customers see live progress in the mobile app.
- View a basic list/history of past orders and simple daily sales overview.

11.3 Supabase Backend

- Central relational database storing users, menu items, orders and order status.
- Authentication service for both customer & staff (Flutter & ASP.NET) accounts.
- Real-time subscriptions so order status changes on the web dashboard reflect instantly in the mobile app, and new orders reflect instantly on the dashboard.
- Storage for menu item images.
- Row-level security rules to ensure customers can only view/edit their own orders and staff-only data is protected.

12. Functional Requirements

ID	Requirement Description	Linked Stakeholder Need	Priority
FR-01	Allow a customer to register and log into the mobile & web app.	Remote/self-service access	High

FR-02	Allow a customer to view the current digital menu with prices.	Browse menu before arriving	High
FR-03	Allow a customer to add items to a cart and submit an order.	Order ahead of arrival	High
FR-04	Allow a customer to view the real-time status of their order.	Reduce uncertainty/queue time	High
FR-05	Allow staff/owner to log into the web & app dashboard securely.	Business data security	High
ID	**Requirement Description**	**Linked Stakeholder Need**	**Priority**
FR-06	Allow staff/owner to add, edit and remove menu items and prices.	Keep menu accurate/current	High
FR-07	Allow staff/owner to view all incoming orders in real time.	Reduce miscommunication	High
FR-08	Allow staff/owner to update the status of an order.	Inform customer of progress	High
FR-09	Notify/update the customer when their order status changes.	Reduce physical waiting	Medium
FR-10	Store completed order history accessible to the customer.	Reorder convenience	Medium
FR-11	Provide the owner with a basic overview/list of past orders.	Business insight/planning	Medium
FR-12	Synchronise order and menu data in real time between the mobile app and the web dashboard via Supabase.	Consistent, accurate information	High

13. Non-Functional Requirements

Category	ID	Requirement Description
Usability	NFR-01	The mobile app shall be simple and intuitive enough for first-time customers with limited technical experience to place an order without assistance.
Performance	NFR-02	The system shall reflect a new order on the staff dashboard within a few seconds of the customer submitting it.
Security	NFR-03	Customer and staff accounts shall be protected using secure authentication, and customers shall only be able to access their own order data.

Availability	NFR-04	The system shall be available during Nama Nameng's normal trading hours, with minimal planned downtime.
Reliability	NFR-05	The system shall accurately save and display order and menu data without data loss under normal operating conditions.
Scalability	NFR-06	The system shall be able to support a growing number of customers and orders as the business's digital reach expands, without major redesign.
Maintainability	NFR-07	The codebase shall be structured and documented so that future developers/students can understand, maintain and extend it.
Compatibility	NFR-08	The Flutter mobile app shall function on common Android and iOS smartphone devices used by the target customer base.

14. System Requirements

14.1 Client-Side (Mobile - Customer)

- Smartphone running Android 8.0+ or iOS 13+.
- Active internet connection (Wi-Fi or mobile data).

14.2 Client-Side (Web - Staff/Owner)

- A device (PC, laptop or tablet) with a modern web browser (Chrome, Edge, Firefox or Safari).
- Active internet connection at the Nama Nameng premises.

14.3 Development Environment

- Flutter SDK and Dart for mobile application development.
- .NET SDK (ASP.NET Core) for the web dashboard.
- Supabase account/project for the backend database, authentication and real-time services.
- Git and a GitHub repository for source control and collaboration.
- Code editor: Visual Studio Code & Visual Studio.

14.4 Server/Backend Requirements

- Cloud-hosted Supabase project (PostgreSQL database, authentication, storage and real-time services).
- Sufficient storage for menu images and order data within Supabase's free/available tier for the prototype.

15. Integration Requirements

The Flutter mobile application, ASP.NET web dashboard and Supabase backend must operate as a single, integrated solution rather than three separate systems.

- The Flutter app shall communicate with Supabase directly (via the Supabase client SDK) for authentication, data reads/writes and real-time order updates.
- The ASP.NET web dashboard shall communicate with Supabase (via REST/API calls or the appropriate .NET client) to manage menu data and retrieve/update orders.

- Both the Flutter app and ASP.NET dashboard shall share the same underlying Supabase database schema for users, menu items and orders, ensuring a single source of truth.
- Real-time subscriptions shall be used so that an order placed on the Flutter app appears immediately on the ASP.NET dashboard, and a status update made on the dashboard is reflected immediately in the Flutter app.
- Authentication tokens issued by Supabase shall be used consistently across both the mobile and web applications to secure access to data.

16. Requirements Prioritisation

The MoSCoW method was used to prioritise the functional requirements listed in Section 12, based on stakeholder input on what is essential to solving the core problem versus what would add value later.

Priority	Requirements
Must Have	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-12 — core registration, ordering, menu management, order visibility and real-time synchronisation.
Should Have	FR-09, FR-10, FR-11 — order notifications, order history and basic sales overview, which improve the experience but are not strictly required for the core problem to be solved.
Could Have	Digital queue position/estimated wait time indicator; ratings/reviews on completed orders.
Won't Have (this version)	Third-party delivery integration, online payment gateway processing, loyalty/rewards programme, multi-branch support (see Section 7.1 Exclusions).

Requirements Change Log

Change ID	Requirement Affected	Change Description	Reason for Change	Date
CL-01	FR-04 / NFR-02	Order status update target relaxed from "instant" to "within a few seconds".	Technical feasibility review found Supabase real-time sync on the free tier is fast but not guaranteed instantaneous.	26 Aug 2026
Change ID	**Requirement Affected**	**Change Description**	**Reason for Change**	**Date**

CL-02	Section 7.1 Exclusions	Online/card payment gateway integration confirmed as fully excluded from this prototype.	Economic and schedule feasibility analysis showed it would add cost, compliance and testing overhead the group cannot absorb this semester.	27 Aug 2026
CL-03	FR-09 (notifications)	In-app status notification confirmed as sufficient; push/SMS notifications excluded.	Schedule feasibility — additional third-party integration and testing time not available.	28 Aug 2026
CL-04	Section 5.1 Scope	Digital queue position/estimated wait time moved from "in scope" to "Could Have".	Risk analysis (R-03) identified this as higher-complexity and a schedule threat if mandatory.	29 Aug 2026

Refined Project Scope

Confirmed In Scope (Baseline for Design)

- Customer registration and login (Flutter mobile app).
- Digital menu browsing with categories, descriptions and prices.
- Order placement (cart and submit) from the mobile app.
- Order status visibility (Received / Preparing / Ready) updated within a few seconds via Supabase realtime sync.
- Owner/staff menu management (add, edit, remove items and prices) via the ASP.NET web dashboard.
- Owner/staff order management — view incoming orders and update order status.
- Basic order history for customers and a simple sales overview for the owner.
- Shared Supabase backend providing the database, authentication and real-time synchronisation.

Confirmed Out of Scope (This Version)

- Online/card payment gateway integration — payment remains cash-on-collection/manual confirmation.
- Third-party delivery platform integration.
- Push/SMS notifications — in-app status updates only.
- Loyalty/rewards programme, advanced analytics, and multi-branch support.

Deferred (Could Have, Time Permitting)

- Digital queue position / estimated wait time indicator.
- Ratings or reviews on completed orders.

Feasibility Study

Technical Feasibility

The group has working knowledge of Flutter, ASP.NET and Supabase from module coursework, all freely available with strong documentation. Real-time synchronisation across three components is the main technical challenge; achievable via Supabase's real-time subscriptions but should be prototyped/spiked early. Conclusion: technically feasible, provided real-time integration is prototyped and validated early.

Operational Feasibility

The solution addresses confirmed pain points and the stakeholder is willing to adopt it. Success depends on owner/staff comfort with the dashboard and customer smartphone access. Conclusion: operationally feasible, provided basic orientation/training is given.

Economic Feasibility

No formal budget; all tools are free and Supabase's free tier suffices for prototype scale. Main cost is the group's time. Conclusion: economically feasible, no material cost to the stakeholder.

Schedule Feasibility

Tight but workable within the semester. MoSCoW prioritisation gives a realistic path; this is the highest-rated risk (R-07) and must be actively monitored. Conclusion: feasible provided Must Have requirements are prioritised and the schedule is followed.

Legal Feasibility

Row-level security (NFR-03) will protect customer data; no unnecessary personal data collected; no credentials uploaded to the public GitHub repo. Conclusion: legally feasible, provided data minimisation and access control are followed.

Ethical Feasibility

Stakeholder participates voluntarily and understands this is a student learning exercise. Conclusion: ethically feasible, provided transparent, respectful engagement continues.

Overall Feasibility Decision

The Nama Nameng project is considered FEASIBLE WITH CHANGES. The core prototype is realistic within the semester timeframe and the group's current skills, provided (1) real-time integration between Flutter, ASP.NET and Supabase is prototyped early, and (2) the group strictly follows MoSCoW prioritisation, descoping Should/Could Have features if necessary.

Project Risk Register

Risks were rated on a 1–5 Likelihood × Impact scale (scores ≥15 = High).

ID	Risk	L	I	Score	Level	Mitigation
R-01	Doja has limited availability due to trading hours, delaying confirmation.	3	3	9	Medium	Schedule meetings around trading hours; use WhatsApp for quick confirmations.
R-02	A group member becomes unavailable/under-contributes.	3	5	15	High	Track tasks weekly via Progress Tracker and GitHub; short weekly check-ins.
R-03	Real-time integration (Flutter/ASP.NET/Supabase) proves more complex than expected.	3	5	15	High	Build a small technical spike of real-time sync early, before full feature development.
R-04	Limited prior experience with Supabase/ASP.NET slows development.	3	3	9	Medium	Allocate dedicated learning time; pair programming; official docs/tutorials.
R-05	Stakeholder requests features beyond agreed scope.	3	3	9	Medium	Communicate confirmed exclusions; confirm scope in writing.
R-06	Unreliable Wi-Fi/data at the premises affects dashboard usability.	2	3	6	Medium	Confirm connectivity during site visits; test with mobile data as backup.
R-07	The semester timeframe is not enough to complete all planned features.	5	5	25	High	Follow MoSCoW strictly; track progress against Microsoft Project weekly.

R-08	Customer/stakeholder data mishandled, stored insecurely, or exposed.	2	5	10	Medium	Apply Supabase RLS; avoid unnecessary personal data; never upload credentials to GitHub.
R-09	Code/documentation lost or overwritten due to poor version control.	2	3	6	Medium	Commit to GitHub regularly with clear messages; use branches for major features.
R-10	Miscommunication within the group causes duplicated/conflicting work.	3	3	9	Medium	Weekly check-ins; shared Progress Tracker; clear task ownership in Microsoft Project.

Three Highest-Rated Risks: R-07 (Score 25) — schedule constraint; R-02 (Score 15) — team member unavailability; R-03 (Score 15) — real-time integration complexity. These are actively monitored at each stakeholder/group checkin.

SWOT Analysis — Nama Nameng Ordering & Queue Management System

	Helps	**Hurts**
Internal	Strengths: solves real stakeholder-confirmed pain points; team knows Flutter/ASP.NET/Supabase; freetier tools; Supabase real-time sync; MoSCoW protects the core deliverable.	Weaknesses: real-time sync across three components is technically complex and unproven; no online payment gateway; very tight semester timeframe (R07); stakeholder access limited to trading hours; no post-module support plan.
External	Opportunities: digital order/sales record opens future stock/staffing decisions; extends reach beyond wordof-mouth; clear path to loyalty/analytics/notifications later; model could adapt to other small food vendors.	Threats: team unavailability (R-02); real-time integration harder than expected (R-03); customer adoption assumes smartphone/data access; Supabase free-tier limits; a competitor could adopt digital ordering first.

17. Appendices: Stakeholder Evidence

This section contains the evidence gathered during stakeholder engagement for Activities 1.1–1.3, confirming that the problem statement, requirements and scope are based on actual stakeholder input rather than group assumptions, as required by the module. Pages are reproduced from the Final Phase 1 SRS.

Appendix A: Interview Notes

SWOT Analysis — Nama Nameng Ordering & Queue Management System

	Helps	Hurts
Internal	Strengths	Weaknesses
	• Solves real, stakeholder-confirmed pain points (queuing time, order-taking errors) — not assumptions • Team already has working knowledge of Flutter, ASP.NET and Supabase • Entirely free-tier tools — no cost to the stakeholder or the team • Supabase real-time sync gives instant visibility between mobile app and web dashboard • MoSCoW prioritisation protects the core deliverable if time runs short	• Real-time sync across three separate components (Flutter, ASP.NET, Supabase) is technically complex and unproven at the time of writing • No online payment gateway in this version — still relies on cash/manual confirmation • Very tight semester timeframe, with almost no schedule slack (this is your highest-rated risk, R-07) • Access to the stakeholder is limited to Nama Nameng's trading hours • It's a prototype — no support or maintenance plan once the module ends
External	Opportunities	Threats
	• A basic digital order/sales record opens the door to future stock and staffing decisions the business can't make today • Extends the business's reach beyond word-of-mouth to new customers • Clear path to add loyalty programmes, analytics, or notifications in a later phase • If it works, the same model could be adapted for other small food vendors in similar situations	• Team member unavailability or under-contribution could delay tasks (R-02) • Real-time integration proving harder than expected could eat into schedule (R-03) • Customer adoption assumes smartphone + mobile data access, which hasn't been fully confirmed • Supabase free-tier limits could be hit if usage grows beyond prototype scale • A competitor could adopt similar digital ordering faster and capture the "convenience" advantage first

17. Appendices: Stakeholder Evidence

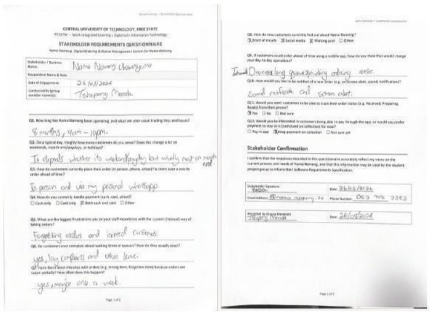
This section will contain the evidence gathered during stakeholder engagement for Activities 1.1–1.3, confirming that the problem statement, requirements and scope above are based on actual stakeholder input rather than group assumptions, as required by the module.

Page 17 of 24

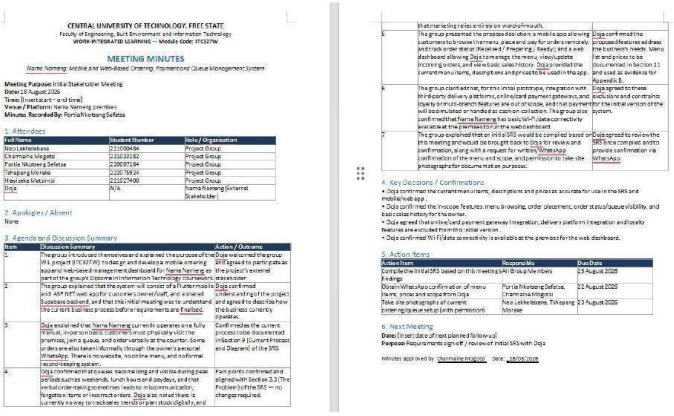
Appendix A — Stakeholder Requirements Questionnaire and Interview Notes

Appendix B: Requirements Confirmation Evidence

Appendix A: Interview Notes



Appendix B: Requirements Confirmation Evidence



CENTRAL UNIVERSITY OF TECHNOLOGY, FREE STATE
Faculty of Engineering, Built Environment and Information Technology
WORK-INTEGRATED LEARNING – Module Code: ITC327W

MEETING MINUTES

Name Naming: Mobile and Web-Based Ordering, Payment and Queue Management System

Meeting Purpose: Initial Stakeholder Meeting
Date: 18 August 2026
Time: [Insert start – end time]
Venue / Platform: Name Naming premises
Minutes Recorded By: Portia Nkobang Sefatsa

1. Attendees

Full Name	Student Number	Role / Organisation
Nazi Lekhelebana	221004494	Project Group
Ckamaine Mogatsi	220120182	Project Group
Portia Nkobang Sefatsa	220007164	Project Group
Tshepang Moriaka	220 159 24	Project Group
Mavleka Matsimbi	221027400	Project Group
Doja	N/A	Name Naming (Internal Stakeholder)

2. Apologies / Absent
None

3. Agenda and Discussion Summary

Item	Discussion Summary	Actions / Outcome
1	The group introduced themselves and explained the purpose of the WIL project (ITC327W) to Doja and developed a mobile ordering application with web-based management dashboard for Name Naming as part of the group's Diploma in Information Technology coursework.	Doja welcomed the group. SRS agreed to participate as the project's external stakeholder.
2	The group explained that the system will consist of a customer-facing mobile app and ASP.NET web app for customers, owner/staff, and a shared business backend, and that this initial meeting was to understand the current business process before requirements are finalised.	Doja confirmed understanding of the project and agreed to describe how the business currently operates.
3	Doja explained that Name Naming currently operates on a fully manual, in-person basis: customers must physically visit the premises, join a queue, and order verbally at the counter. Some orders are also taken informally through the owner's personal WhatsApp. There is no website, no online menu, and no formal record-keeping system.	Confirmed as the current process to be documented in Section 2 (Current Process and Diagram) of the SRS.
4	Doja confirmed that classes back-to-back and visible during peak periods such as weekends, lunch hours and paydays, and that verbal order-taking sometimes leads to miscommunication, forgotten items or incorrect orders. Doja also noted that there is currently no way to track sales trends or plan stock digitally, and that marketing relies entirely on word-of-mouth.	Pain points confirmed and aligned with Section 3.3 (The Problem) of the SRS — no changes required.
5	The group presented the proposed solution: a mobile app allowing customers to browse the menu, place and pay for orders remotely, and track order status (Received / Preparing / Ready); and a web dashboard allowing Doja to manage the menu, view/update incoming orders, and view basic sales history. Doja provided the current menu items, descriptions and prices to be used in the app.	Doja confirmed the proposed features address the customer needs. Menu list and prices to be documented in Section 11 and used as evidence for Appendix B.
6	The group clarified that, for this initial prototype, integration with third-party delivery platforms, online/card payment gateways, and loyalty or multi-branch features are out of scope, and that payment will be simulated or handled as cash on collection. The group also confirmed that Name Naming has basic Wi-Fi/data connectivity available at the premises for the web dashboard.	Doja agreed to these exclusions and constraints for the initial version of the system.
7	The group explained that an initial SRS would be compiled based on this meeting and would be brought back to Doja for review and confirmation, along with a request for written WhatsApp confirmation of the menu and scope, and permission to take site photographs for documentation purposes.	Doja agreed to review the SRS once compiled and to provide confirmation via WhatsApp.

4. Key Decisions / Confirmations
- Doja confirmed the current menu items, descriptions and prices as accurate for use in the SRS and mobile/web app.
- Doja confirmed the in-scope features: menu browsing, order placement, order status/queue visibility, and basic order history for the customer.
- Doja agreed that online/card payment gateway integration, delivery platform integration and loyalty features are excluded from this initial version.
- Doja confirmed Wi-Fi/data connectivity is available at the premises for the web dashboard.

5. Action Items

Action Item	Responsible	Due Date
Compile the Initial SRS based on this meeting and Doja's feedback	Portia Nkobang Sefatsa (Name Naming Project Findings)	23 August 2026
Obtain WhatsApp confirmation of menu items, prices and scope from Doja	Portia Nkobang Sefatsa, Ckamaine Mogatsi	22 August 2026
Take site photographs of current operations (with permission)	Nazi Lekhelebana, Tshepang Moriaka	23 August 2026

6. Next Meeting
Date: [Insert date of next planned follow-up]
Purpose: Requirements sign-off / review of Initial SRS with Doja

Minutes approved by: Ckamaine Mogatsi Date: 18/08/2026

Appendix B — Meeting Minutes, Initial Stakeholder Meeting (18 August 2026)

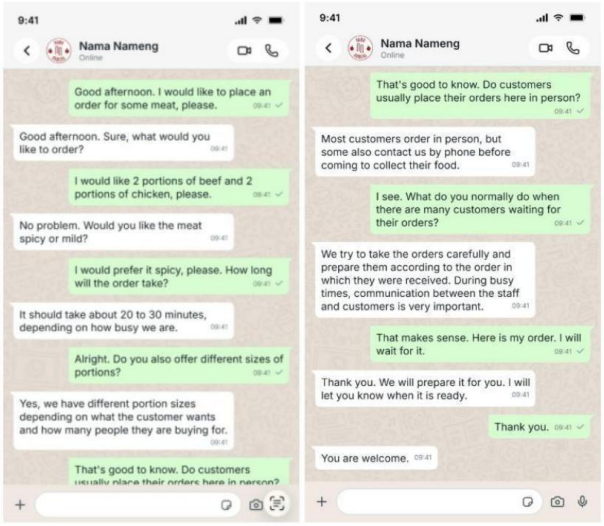
Appendix C: Site Visit / Photographic Evidence



Appendix C — Site visit photographs at Nama Nameng premises

Appendix D: Communication Log (Phase 1)

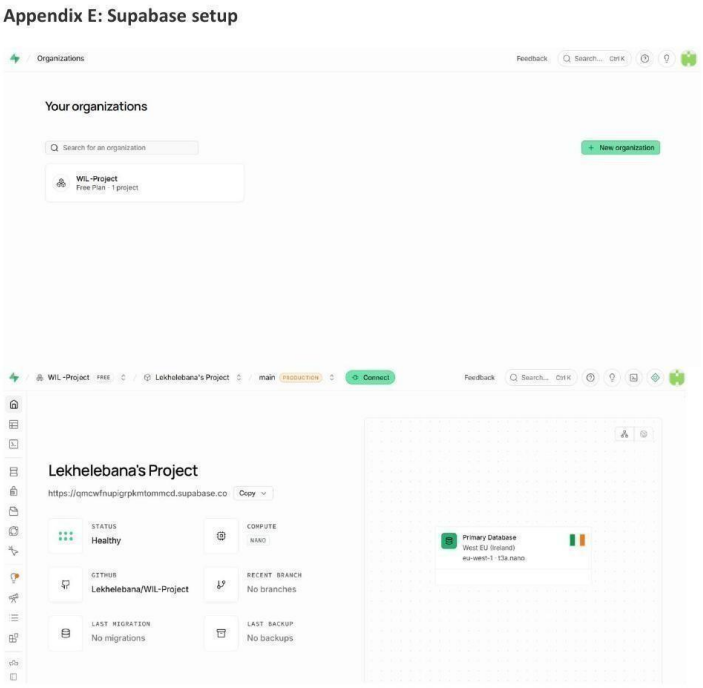
Appendix D: Communication Log



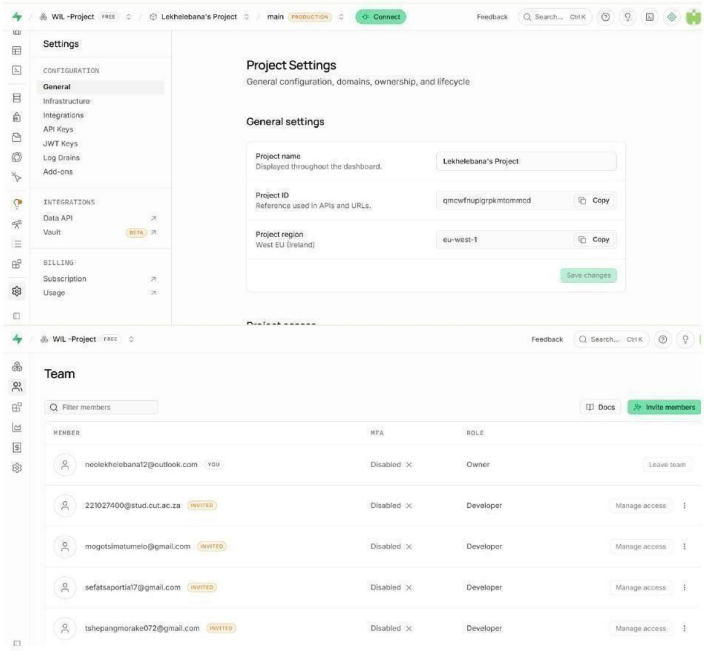
Appendix D — WhatsApp communication log with Doja (Phase 1). See Section 18.4 / Section 24 of this document: this Appendix D — WhatsApp communication log with Doja (Phase 1).

The Phase 2 design review is recorded in Section 25.appendix still needs a Phase 2 design-review counterpart.

Appendix E: Supabase Setup

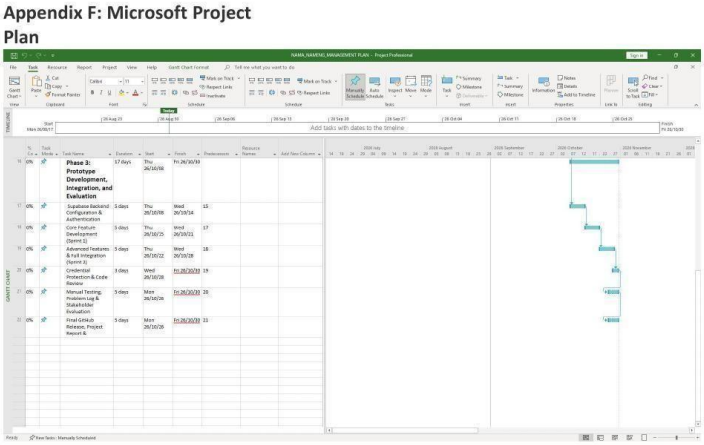


Appendix E — Supabase project setup, page 1

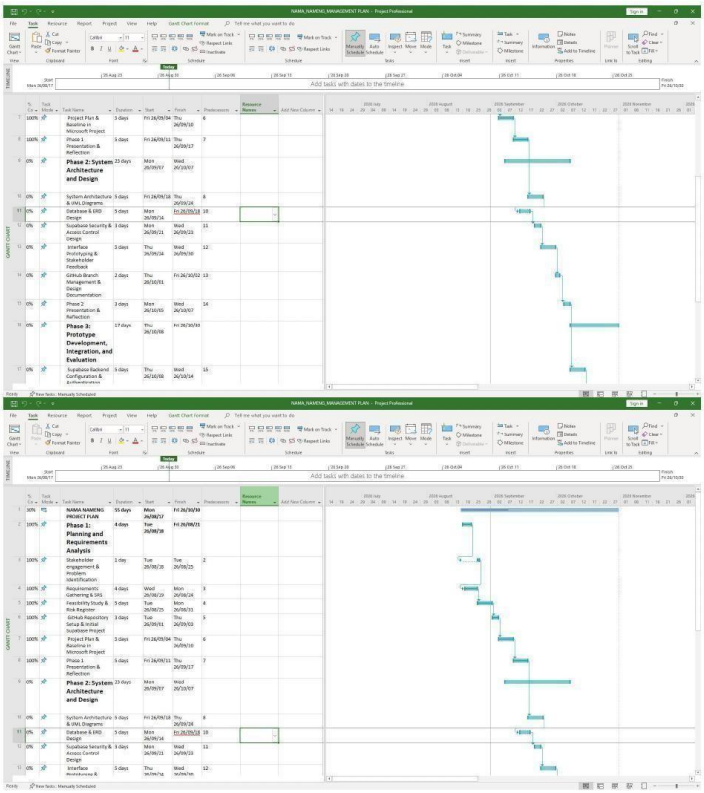


Appendix E — Supabase project setup, page 2 (Team access)

Appendix F: Microsoft Project Plan



Appendix F — Microsoft Project Gantt chart, Phase 3



Appendix G: Github Repository Link

https://github.com/Lekhelebana/WIL-Project

Appendix F — Microsoft Project Gantt chart, Phases 1–2

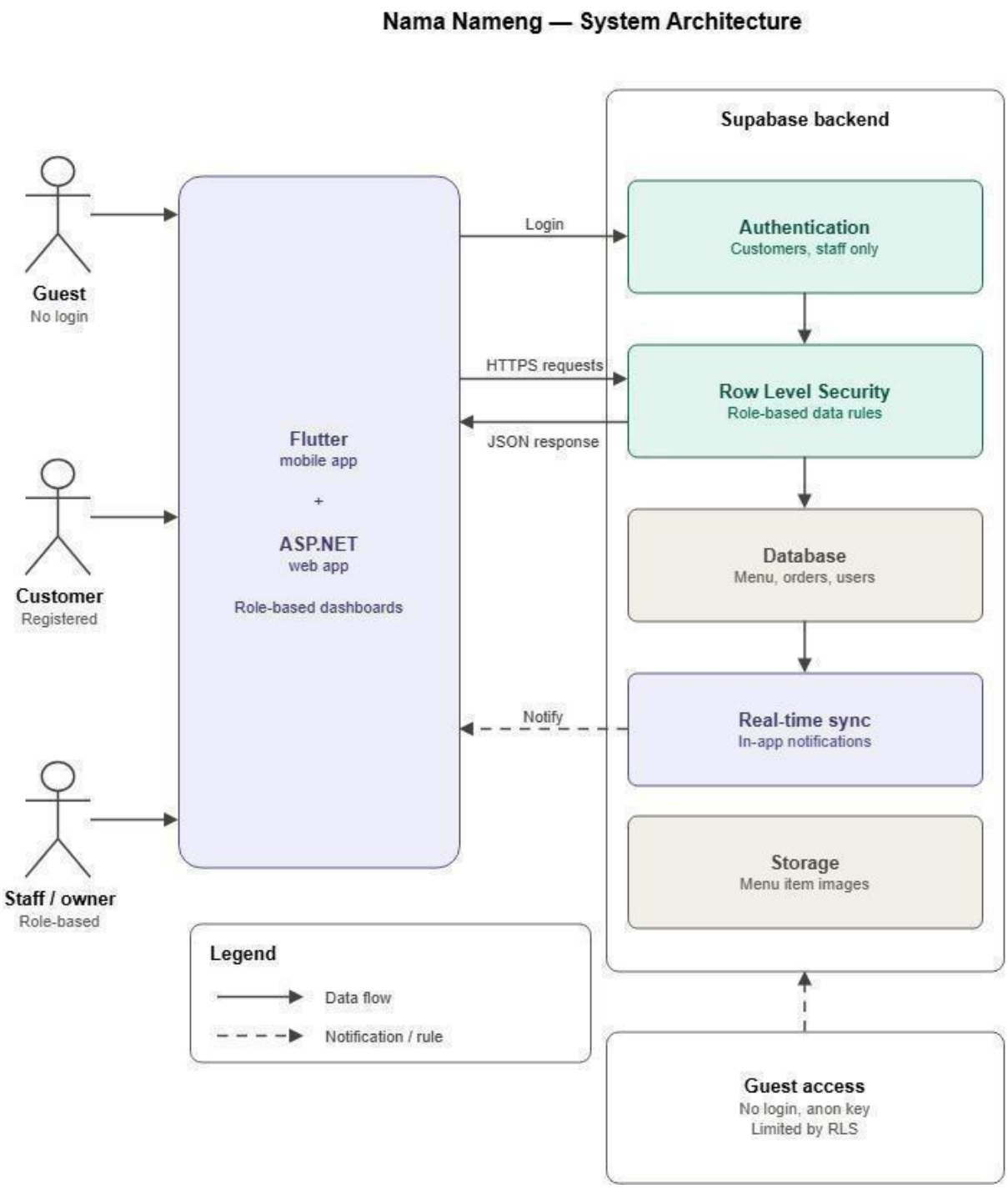
Appendix G: GitHub Repository Link

https://github.com/Lekhelebana/WIL-Project

18. Phase 2: System Architecture and Design

18.1. System Architecture Diagram

Flutter, ASP.NET and Supabase as one integrated system, with relevant users, connections and data flows.



18.2 Use Case Diagram

[UseCaseDiagram.drawio - draw.io]

The Use Case Diagram below identifies four actors — Guest, Customer, Staff and Manager — drawn directly from the actors already used in the four sequence diagrams. Two generalisation relationships are used, both reflecting real behaviour confirmed in the sequence diagrams rather than notation for its own sake: Customer generalises Guest, because the Guest sequence diagram shows that browsing, ordering and receiving status updates work identically whether or not the person registers — Customer simply adds Register/Login and order history on top of what a Guest can already do. Manager generalises Staff for the same reason: the Manager sequence diagram explicitly inherits every Staff use case and adds only menu management and the sales overview.

Two <<include>> relationships are used because the included behaviour always executes as part of the base use case: Place Order always includes Browse Digital Menu and Add to Cart (an order cannot exist without them, per the Guest and Customer sequence diagrams), and Update Order Status always includes Notify Customer (every status save is immediately followed by a push notification in all four sequence diagrams). Two <<extend>> relationships are used for Register Account and Login against Place Order, because the sequence diagrams state registration and login are optional add-ons available at any point during ordering, not mandatory steps.

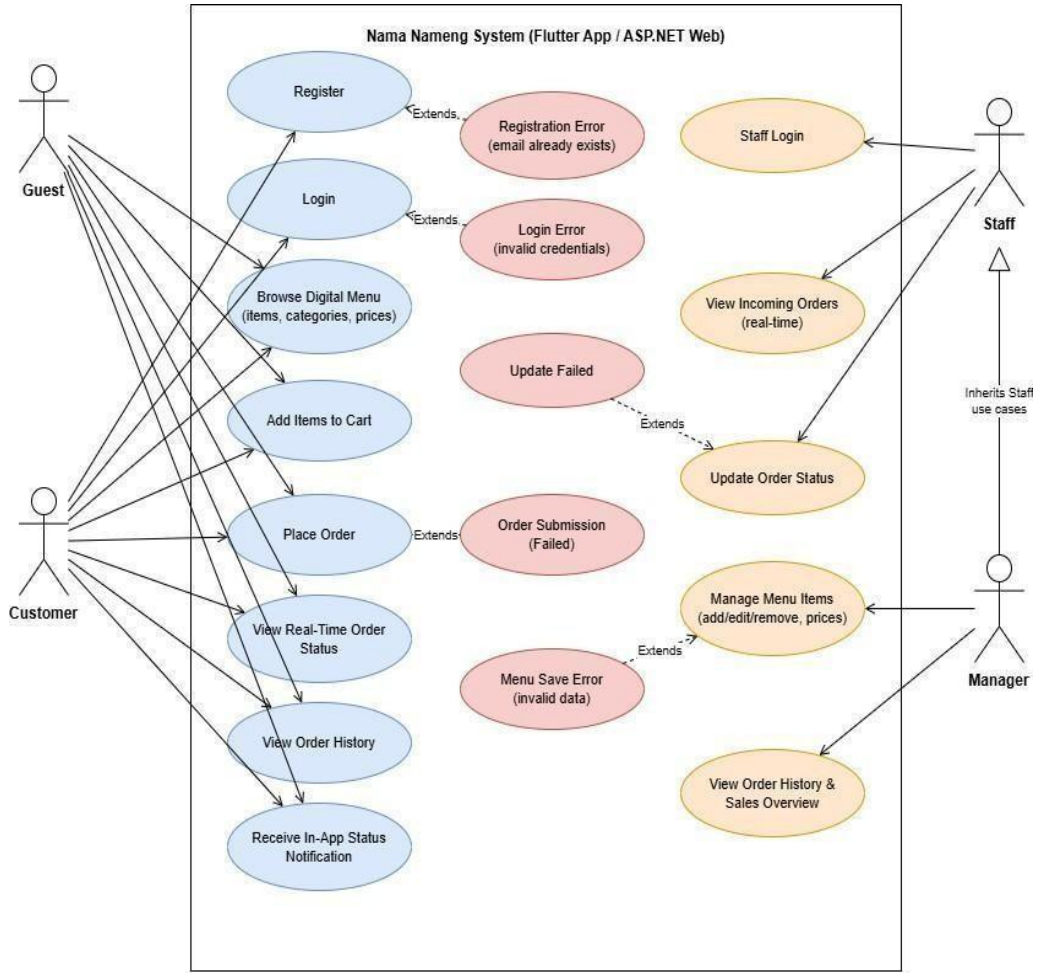
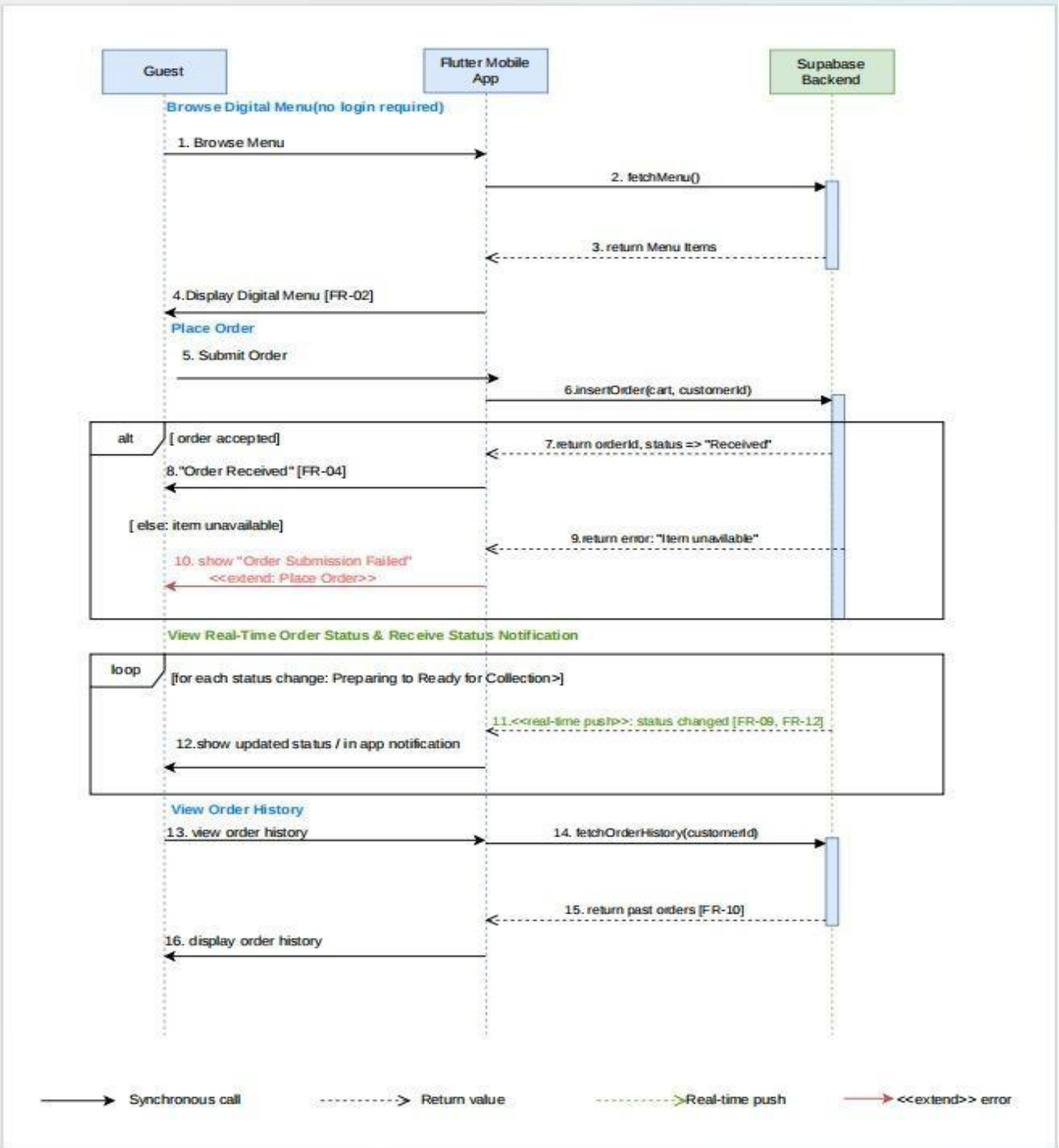


Figure 18.1: Nama Nameng Use Case Diagram

19. Sequence Diagrams

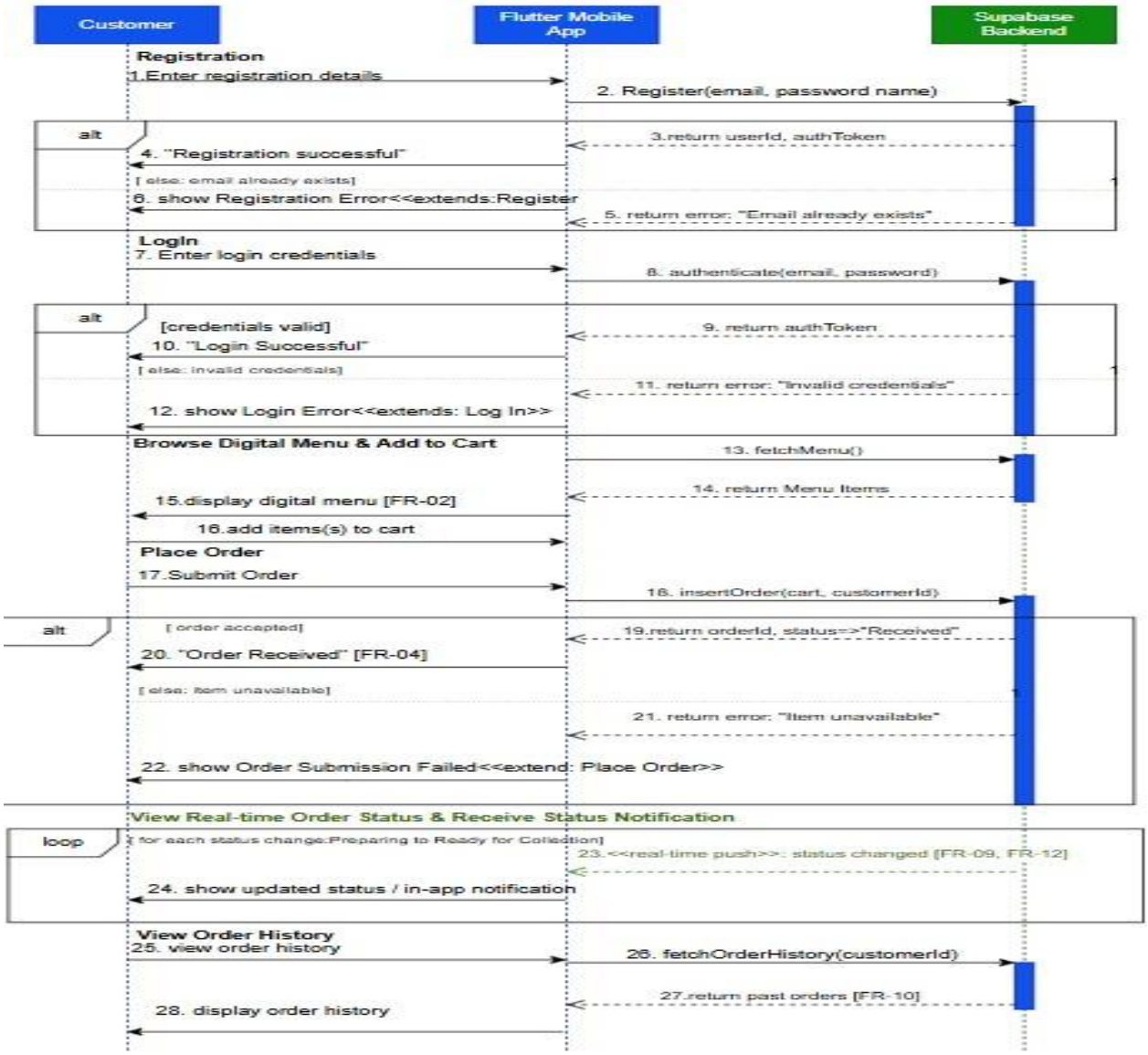
The Phase 2 addendum and Project Progress Tracker (Task U2-03) reference four sequence diagrams — Guest, Customer, Staff and Manager. One representing an important Flutter process and one an important ASP.NET process, with clear lifelines, correctly ordered messages, and each matching a use case from Section 18.1.

19.1 Guest Sequence Diagram



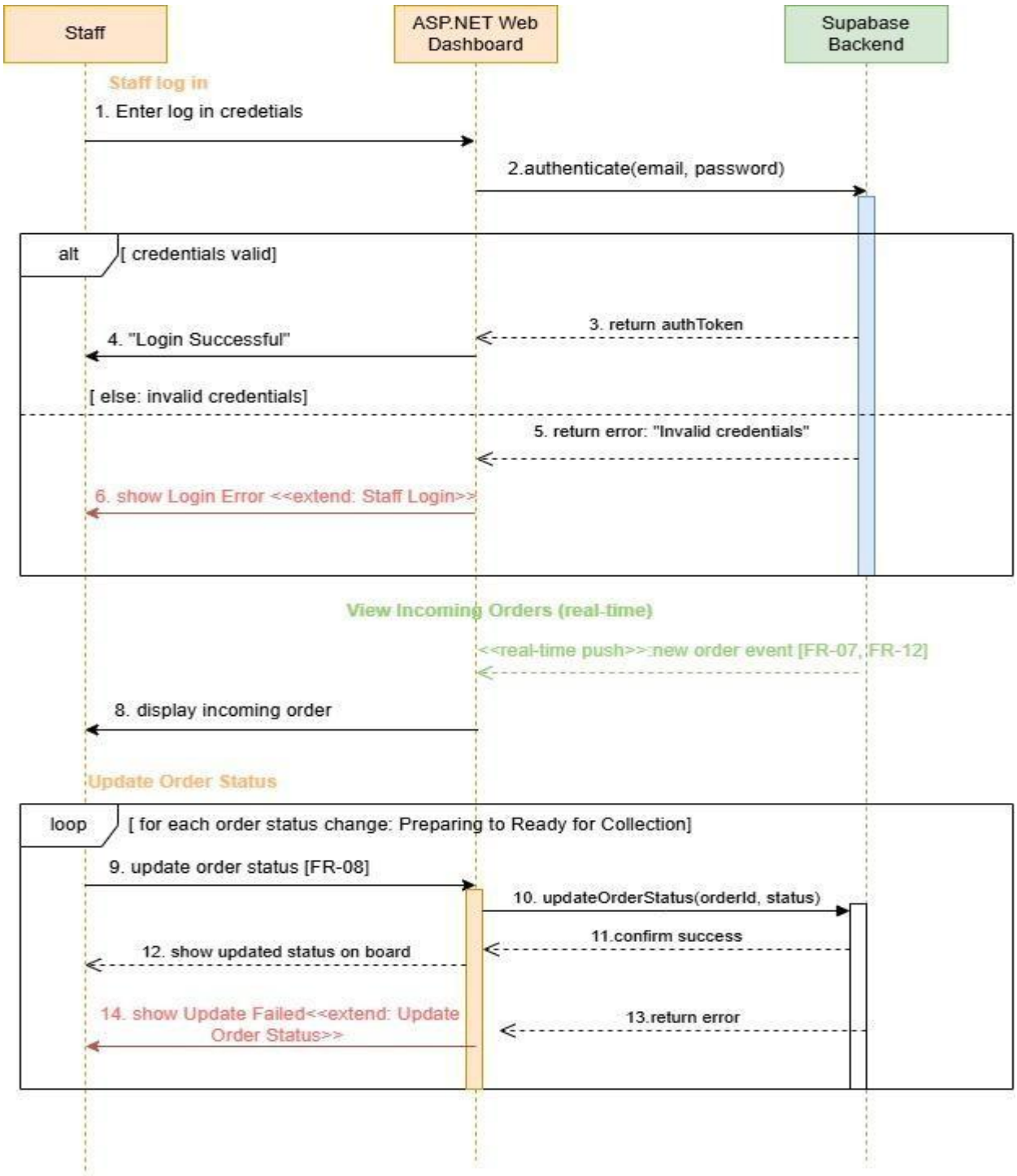
19.2 Customer Sequence Diagram

[CustomerSequenceDiagram.drawio - draw.io]



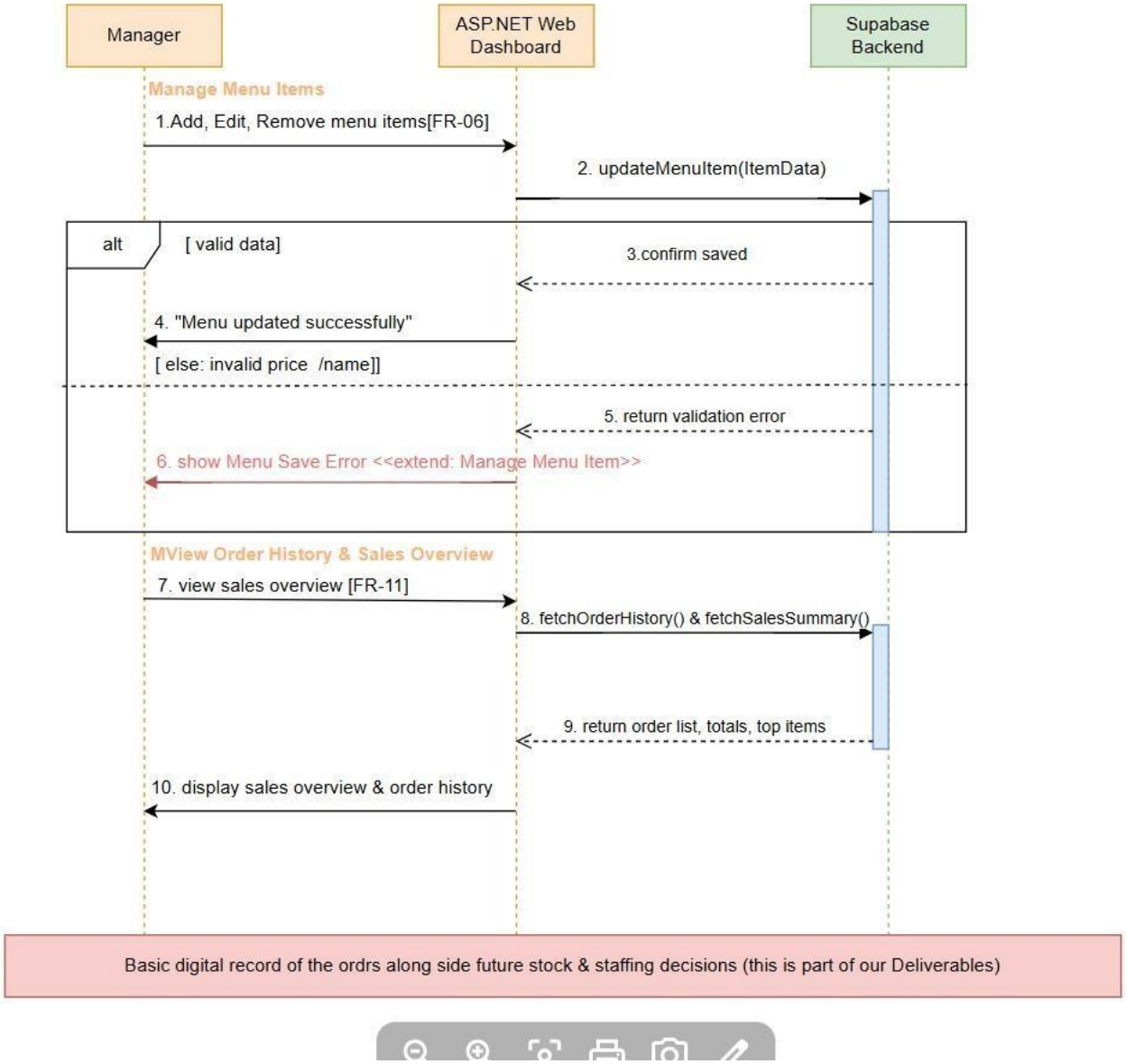
19.3 Staff Sequence Diagram

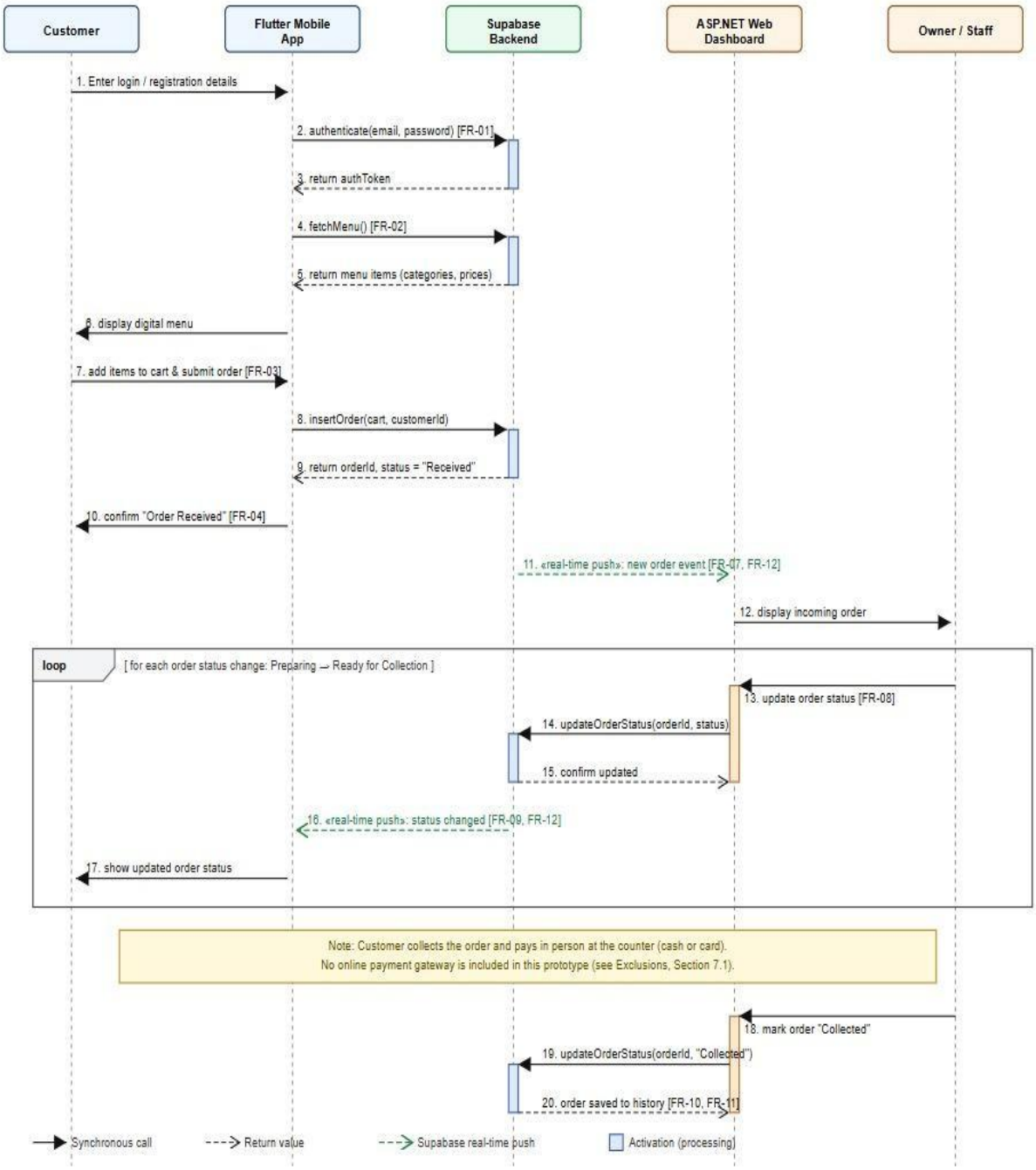
[StaffSequence.drawio - draw.io]



19.4 Manager Sequence Diagram

[ManagerSequence.drawio - draw.io]

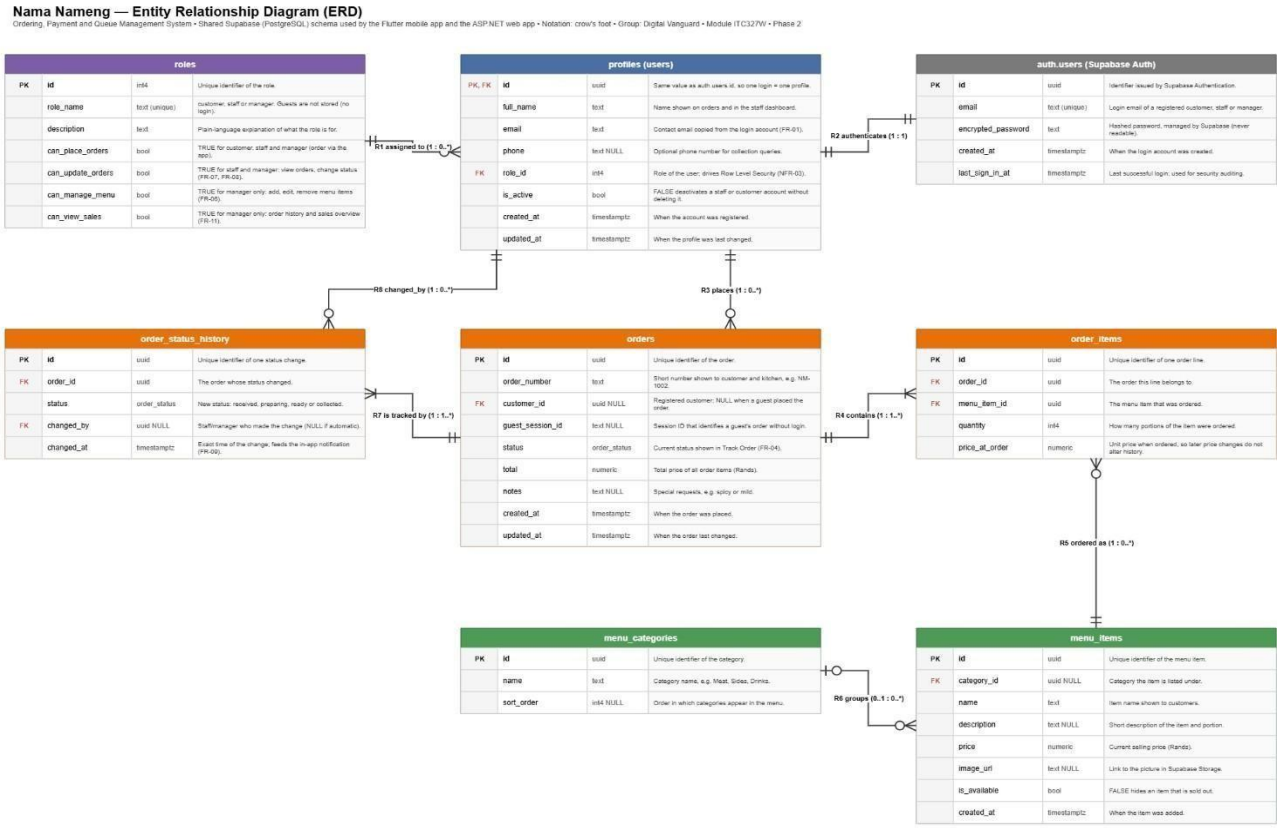

Manager
ASP.NET Web Dashboard
Supabase Backend
Manage Menu Items
1.Add, Edit, Remove menu items[FR-06]
2. updateMenuItem(ItemData)
alt
[valid data]
3.confirm saved
4. "Menu updated successfully"
[else: invalid price /name]]
5. return validation error
6. show Menu Save Error <<extend: Manage Menu Item>>
MView Order History & Sales Overview
7. view sales overview [FR-11]
8. fetchOrderHistory() & fetchSalesSummary()
9. return order list, totals, top items
10. display sales overview & order history
Basic digital record of the ordrs along side future stock & staffing decisions (this is part of our Deliverables)


Customer
Flutter Mobile App
Supabase Backend
ASP.NET Web Dashboard
Owner / Staff
1. Enter login / registration details
2. authenticate(email, password) [FR-01]
3. return authToken
4. fetchMenu() [FR-02]
5. return menu items (categories, prices)
6. display digital menu
7. add items to cart & submit order [FR-03]
8. insertOrder(cart, customerId)
9. return orderId, status = "Received"
10. confirm "Order Received" [FR-04]
11. «real-time push»: new order event [FR-07, FR-12]
12. display incoming order
loop
[for each order status change: Preparing → Ready for Collection]
13. update order status [FR-08]
14. updateOrderStatus(orderId, status)
15. confirm updated
16. «real-time push»: status changed [FR-09, FR-12]
17. show updated order status
Note: Customer collects the order and pays in person at the counter (cash or card).
No online payment gateway is included in this prototype (see Exclusions, Section 7.1).
18. mark order "Collected"
19. updateOrderStatus(orderId, "Collected")
20. order saved to history [FR-10, FR-11]
Synchronous call
Return value
Supabase real-time push
Activation (processing)

20. Entity Relationship Diagram (ERD)

The checkpoint requires entities/tables, important attributes/fields, primary keys, foreign keys, relationships and correct cardinality, supporting the data referenced throughout Section 18.2 (profiles, menu_items, menu_categories, orders, order_items, order_status_history).

https://app.diagrams.net/#G1CJb0Ckp6RlVyDebzuBgKXC8RBG-dN6qF#%7B%22pageId%22%3A%22nama-nameng-erd%22%7D



Roles and user types

Actor	Stored in database?	Login	What they can do	Related requirements
Guest	No user row. Orders carry guest_session_id.	None (Supabase anon key, limited by RLS)	Browse the menu, add to cart, place an order, see the status of their own order. Registering is optional.	FR-02, FR-03, FR-04
Customer (role: customer)	auth.users + profiles + roles	Email and password	Everything a guest can do, plus saved order history and profile. Can only read their own orders.	FR-01, FR-10, NFR-03
Staff (role: staff)	auth.users + profiles + roles	Email and password (Portal Login)	See incoming orders in real time and update order status. Cannot change the menu or see sales.	FR-05, FR-07, FR-08
Manager / Owner (role: manager)	auth.users + profiles + roles	Email and password (Portal Login)	All staff functions plus add/edit/remove menu items and categories and view order history and sales overview.	FR-06, FR-11

Legend and enumerated types

PK = primary key • **FK** = foreign key • **NULL** = value is optional
1 = exactly one • **0..1** = zero or one • **0..*** = zero or many • **1..*** = one or many

order_status: received, preparing, ready, collected
Colour code: purple = access control, blue = users, grey = Supabase Auth, orange = orders, green = menu

Design notes

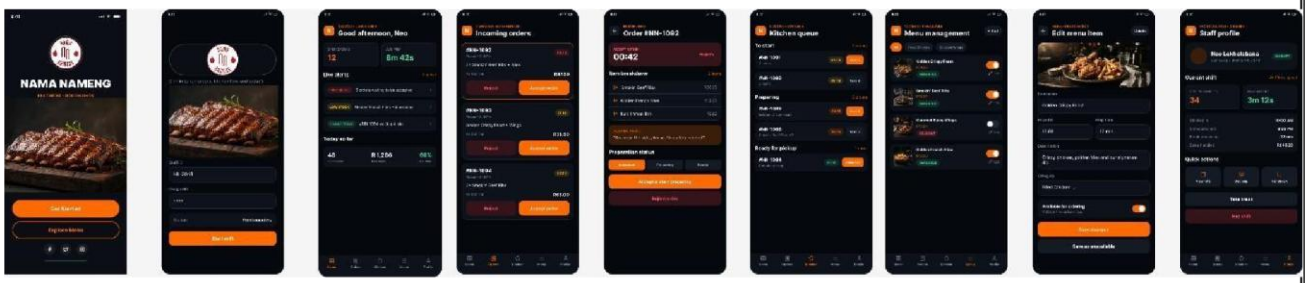
- **Guest ordering:** customer_id may be NULL and guest_session_id identifies the guest (Guest sequence diagram).
- **Security:** Row Level Security uses profiles.role_id and the roles permission columns so customers only see their own orders (NFR-03).
- **Real time:** Supabase Realtime watches orders and order_status_history for live dashboards and in-app notifications (FR-12).
- **Price history:** price_at_order protects old orders and sales figures when menu prices change.
- **Payment and cart:** payment is cash on collection and the cart is held in the app until Place Order, so neither has a table (Section 7.1).

21. Flutter Interface Designs

Main mobile screens for the customer-facing app and the staff/manager app, matching the screens named in Section 18.2 (Welcome Back / Getting Started, Menu / Home, Checkout Summary, Track Order, Profile / order history).

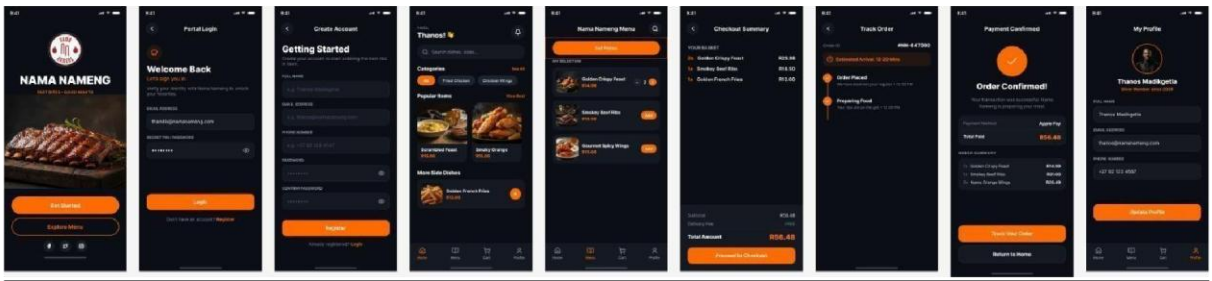
STAFF/MANAGER FLUTTER APP

https://www.figma.com/design/tVj3ozN921e90wa2N6ZQO5/Staff-App?node-id=01&p=f&t=OpOHL8aeV4anfivv-0



CUSTOMER/GUEST FLUTTER APP

https://www.figma.com/design/5N5aljGZvRRWxidTQu8vF2/Final-Flutter-Customer?node-id=0-1&t=anBYrWINvvFZuD5g-1



22. ASP.NET Interface Designs

Main web screens for the customer-facing web app and the staff/owner dashboard, matching the screens named in Section 18.2 (Portal Login, ASP.NET menu management, ASP.NET orders dashboard, ASP.NET order detail, ASP.NET sales overview).

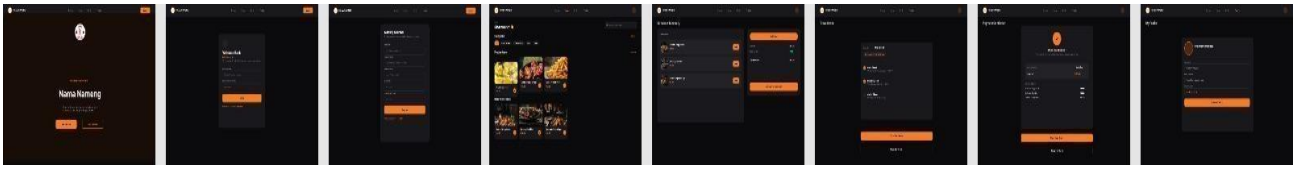
STAFF/MANAGER WEB APP

https://www.figma.com/design/kEfxKbTQmIpcMNZVW7iUfO/ASP.NET-staff?node-id=0-1&p=f

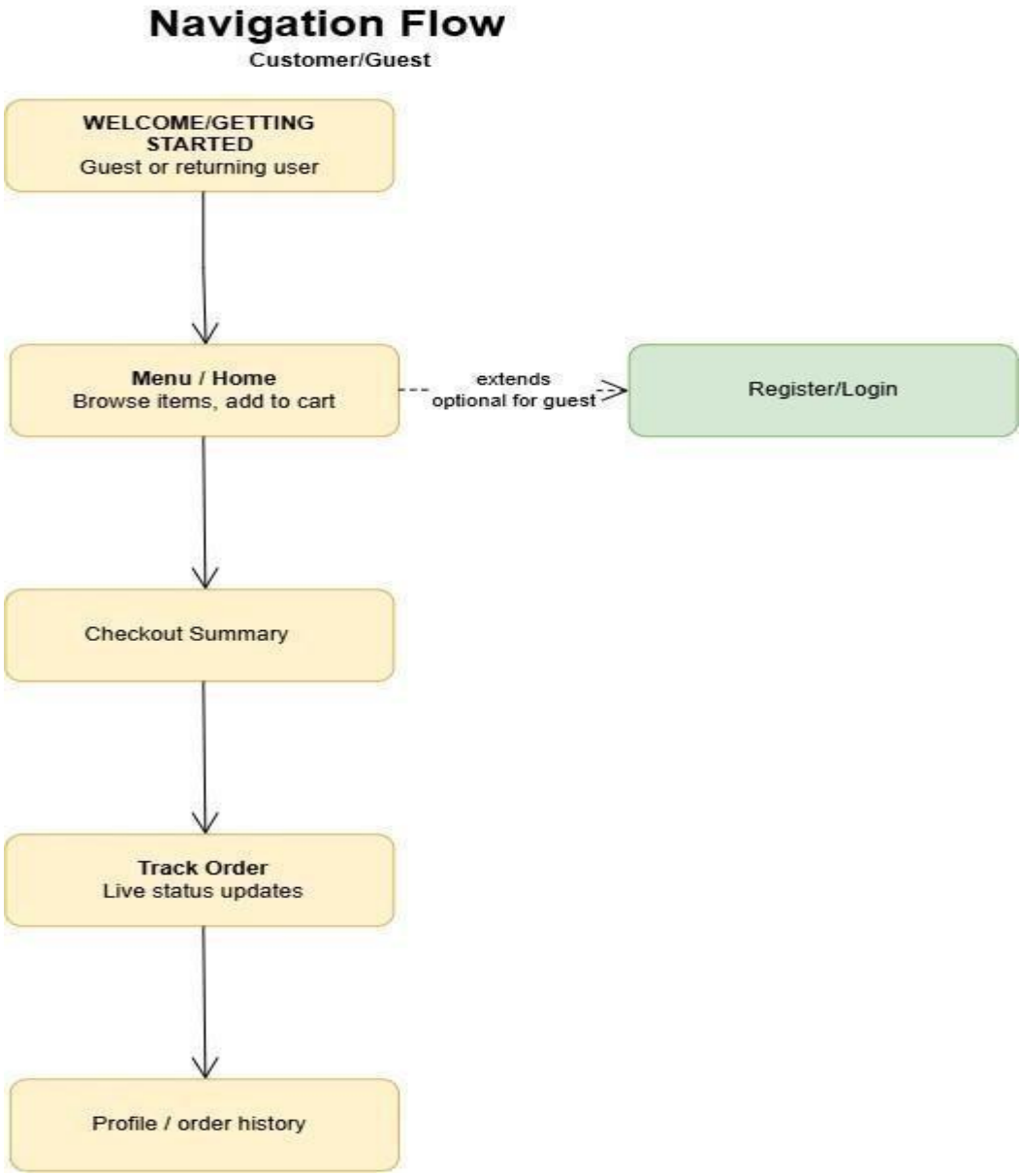


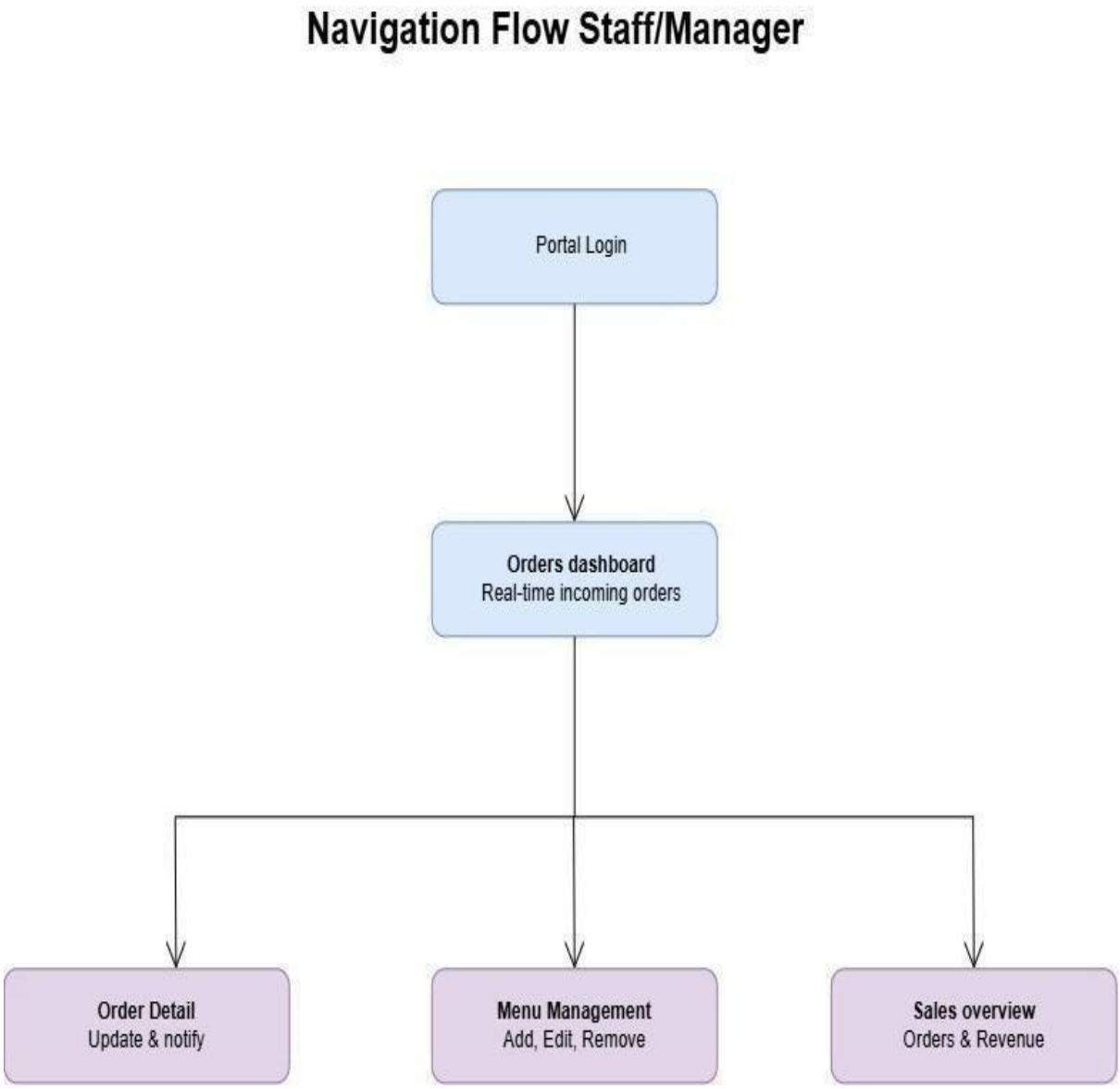
CUSTOMER/GUEST WEB APP

https://www.figma.com/design/amXSFjyoYmmCv5mB5wU4E9/Customer-Web-App?node-id=01&t=qrazwvro3H4UkpON-0



24 Navigation Flow Diagram





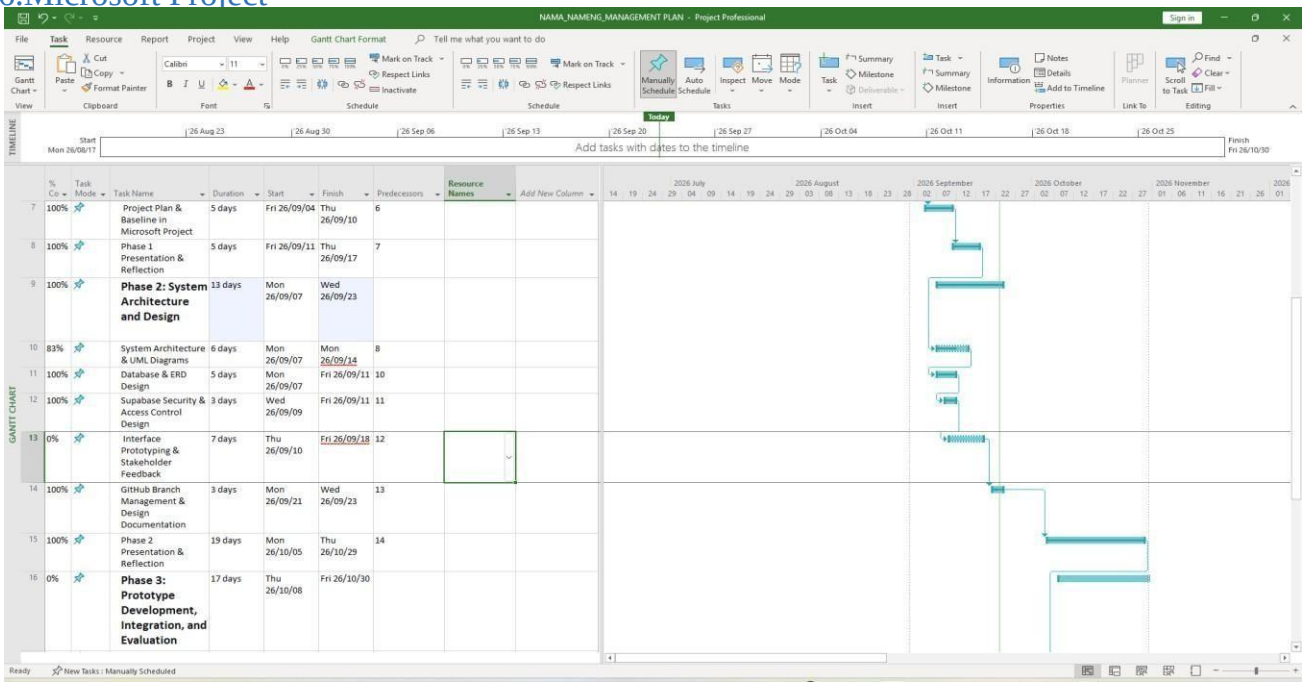
25. Phase 2 Stakeholder Design Review

Field	Details
Date and attendees	20 September 2026 All members were present
What was shown to Doja	System Architecture Diagram, Use Case Diagram, Interface Screens

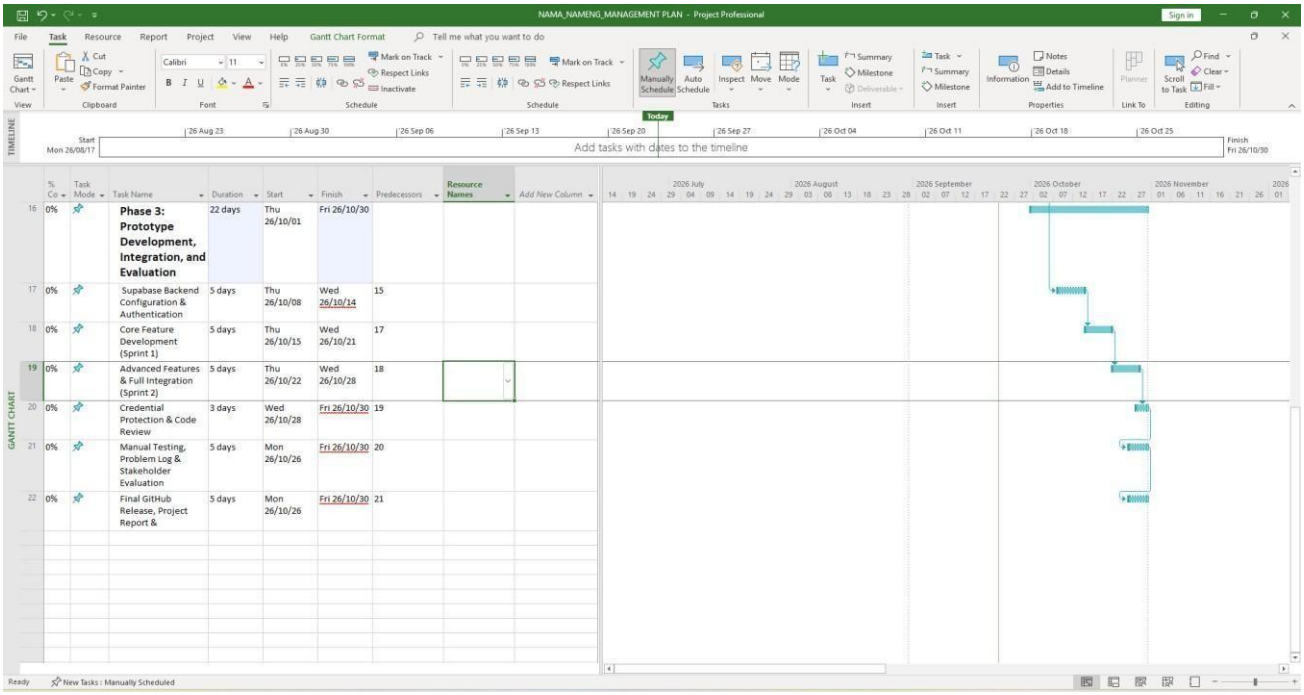
Doja's feedback	Doja was pleased with the progress made and is eager to see the final product.
Changes requested	N/A
Changes made in response	N/A



26.Microsoft Project



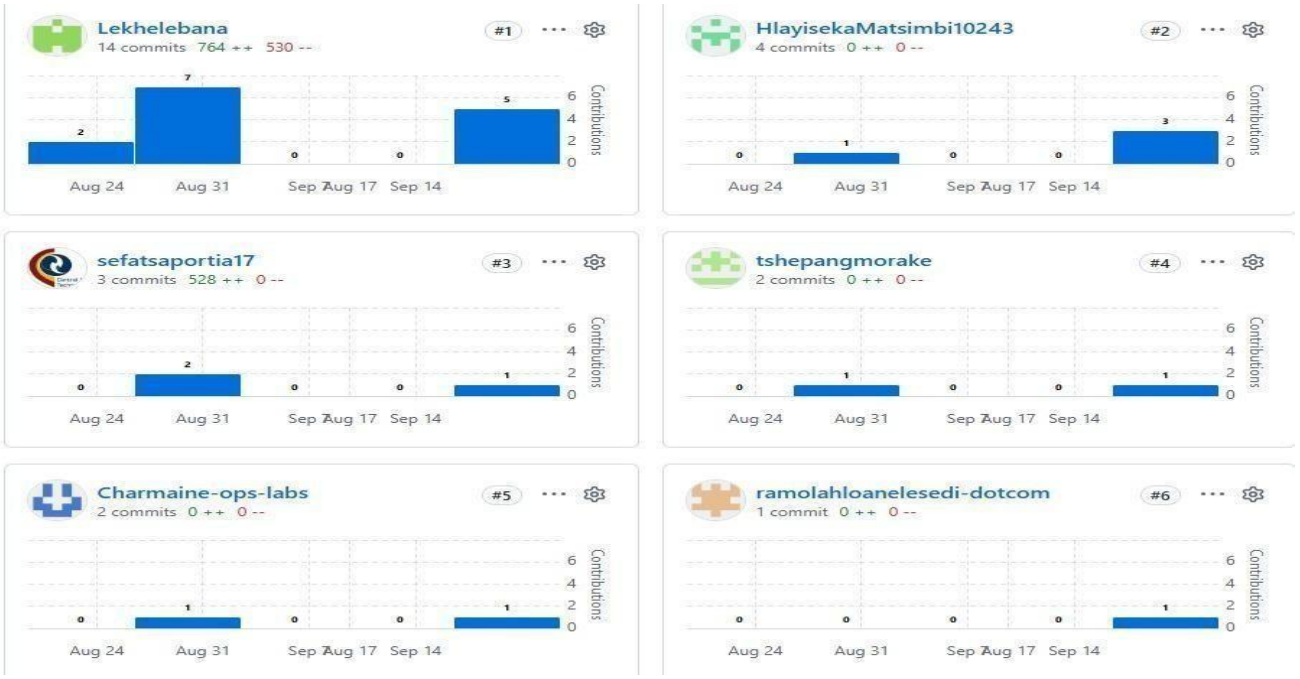
	% Co	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
7	100%		Project Plan & Baseline in Microsoft Project	5 days	Fri 26/09/04	Thu 26/09/10	6	
8	100%		Phase 1 Presentation & Reflection	5 days	Fri 26/09/11	Thu 26/09/17	7	
9	100%		**Phase 2: System Architecture and Design**	13 days	Mon 26/09/07	Wed 26/09/23		
10	83%		System Architecture & UML Diagrams	6 days	Mon 26/09/07	Mon 26/09/14	8	
11	100%		Database & ERD Design	5 days	Mon 26/09/07	Fri 26/09/11	10	
12	100%		Supabase Security & Access Control Design	3 days	Wed 26/09/09	Fri 26/09/11	11	
13	0%		Interface Prototyping & Stakeholder Feedback	7 days	Thu 26/09/10	Fri 26/09/18	12	
14	100%		GitHub Branch Management & Design Documentation	3 days	Mon 26/09/21	Wed 26/09/23	13	
15	100%		Phase 2 Presentation & Reflection	19 days	Mon 26/10/05	Thu 26/10/29	14	
16	0%		**Phase 3: Prototype Development, Integration, and Evaluation**	17 days	Thu 26/10/08	Fri 26/10/30		

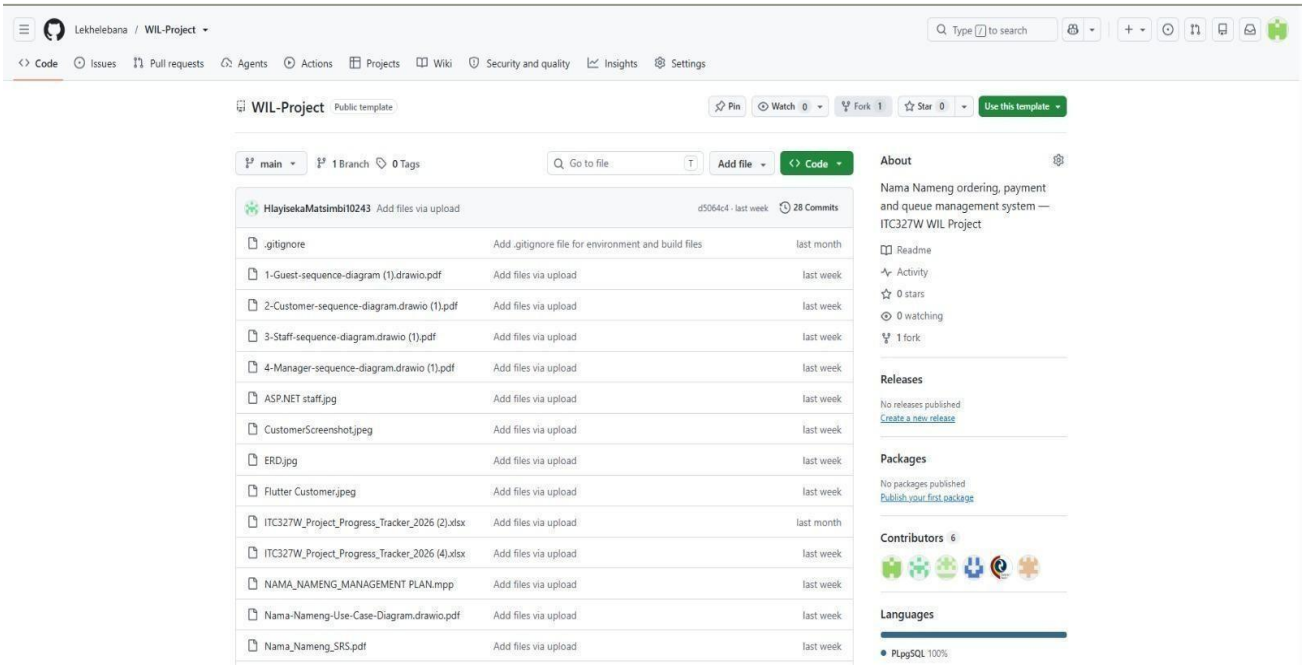


	% Co	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
16	0%		**Phase 3: Prototype Development, Integration, and Evaluation**	22 days	Thu 26/10/01	Fri 26/10/30		
17	0%		Supabase Backend Configuration & Authentication	5 days	Thu 26/10/08	Wed 26/10/14	15	
18	0%		Core Feature Development (Sprint 1)	5 days	Thu 26/10/15	Wed 26/10/21	17	
19	0%		Advanced Features & Full Integration (Sprint 2)	5 days	Thu 26/10/22	Wed 26/10/28	18	
20	0%		Credential Protection & Code Review	3 days	Wed 26/10/28	Fri 26/10/30	19	
21	0%		Manual Testing, Problem Log & Stakeholder Evaluation	5 days	Mon 26/10/26	Fri 26/10/30	20	
22	0%		Final GitHub Release, Project Report &	5 days	Mon 26/10/26	Fri 26/10/30	21	

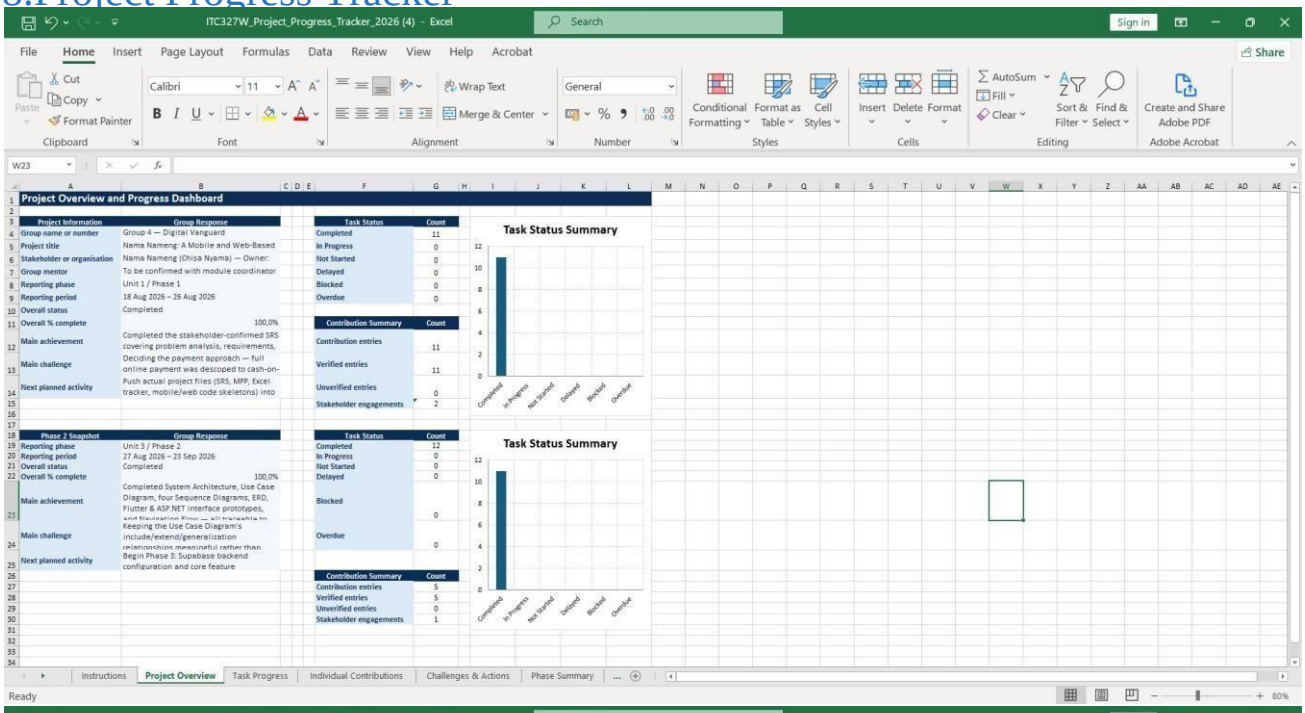
27.GITHUB

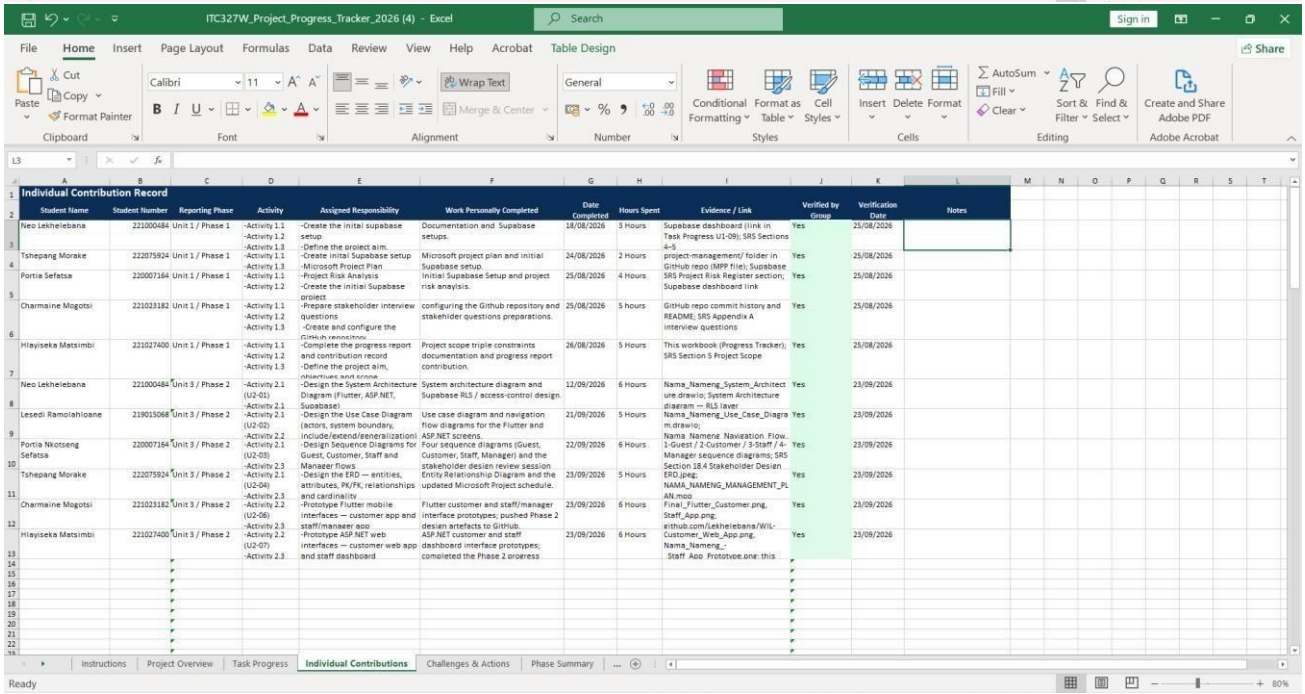
github.com/Lekhelebana/WIL-Project





28.Project Progress Tracker





29. Phase 3: Implementation and Final Prototype

This section continues the SRS from Phases 1 and 2. It records what was built in Phase 3: the Flutter mobile application, the ASP.NET web application and the shared Supabase backend, delivered as one integrated prototype for Nama Nameng. It links the approved requirements to the implemented functions, describes the shared processes between the two applications, lists the changes approved during implementation and states the known limitations.

29.1 Overview of Implemented Components

Component	What was implemented	Technology
Shared Supabase backend	Six tables (profiles, menu_categories, menu_items, orders, order_items, order_status_history) with primary and foreign keys; three helper functions (is_staff, is_manager and the sign-up trigger handle_new_user); two order functions (place_order and set_order_status); row-level security on every table; a Realtime publication on orders. Delivered as four SQL scripts: schema, menu seed data, staff setup and a column fix.	PostgreSQL, Supabase Auth, row-level security, Realtime
Flutter mobile application	One app with two role-based areas: a customer area (menu, cart, checkout, order tracking, history) and a staff/manager area (incoming orders, kitchen queue, menu management). About 3,100 lines of Dart and 7 unit tests.	Flutter, Dart, supabase_flutter
ASP.NET web application	MVC application with a customer site and a staff/manager portal on the same Supabase project. About 900 lines of controllers plus Razor views. Cookie sessions; staff and manager routes restricted by role.	ASP.NET Core 8 MVC, Razor, EF Core with Npgsql

29.2 Shared Supabase Backend

Both applications use one Supabase project and the same records. The database follows the approved ERDOrder status values are Received, Preparing, Ready, Collected and Cancelled.

Table	Main fields	Keys and relationships	Purpose
profiles	id, full_name, email, phone, role (Customer, Staff or Manager), is_active, created_at, updated_at	PK id; linked one-to-one to the Supabase Auth user	Profile and role for every signed-in user
menu_categories	id, name, sort_order	PK id	Menu groupings shown to customers
menu_items	id, category_id, name, description, price, image_url, is_available, created_at	PK id; FK category_id to menu_categories	Menu with prices and a sold-out flag
orders	id, order_number (NN-1001 upward), customer_id, guest_session_id, status, total, notes, created_at, updated_at	PK id; FK customer_id to profiles (empty for guests)	One row per order; guests are identified by guest_session_id
order_items	id, order_id, menu_item_id, quantity, price_at_order	PK id; FK order_id to orders; FK menu_item_id to menu_items	Order lines; price is frozen at order time

Table	Main fields	Keys and relationships	Purpose
order_status_history	id, order_id, status, changed_by, changed_at	PK id; FK order_id to orders; FK changed_by to profiles	Record of every status change

Permitted operations and how they are enforced:

Operation	Who may do it	How it is enforced
Read the menu	Everyone, including guests	Public read policies on menu_categories and menu_items
Place an order	Guest or customer	place_order() calculates prices on the server, creates the order, its lines and the first history row
Read orders	A customer sees only their own orders; staff and manager see all	orders_select policy (Flutter); role checks in the controllers (ASP.NET)
Update order status	Staff and manager	set_order_status() checks the role and writes a history row
Add, edit, remove menu items	Manager only	items_write policy using is_manager() (Flutter); manager-only controllers (ASP.NET)

How each front end uses the shared project:

Component	How it talks to Supabase	Sign-in	Live updates
Flutter app	supabase_flutter client and RPC calls; row-level security applies	Email and password; anonymous sign-in for guests	Realtime subscription plus a 5-second timer as fallback
ASP.NET web	EF Core to Postgres through the Supabase session pooler; Supabase Auth over HTTPS; cookie session	The same Supabase accounts; session-based guest id	Page refresh by polling every 4 to 5 seconds
Supabase	PostgreSQL, Auth, row-level security, Realtime	Roles stored on profiles	Publication on orders

29.3 Flutter Mobile Application

The Flutter application follows the approved interface designs (Section 21) and navigation flows (Section 24). Guests can order without registering by using Supabase anonymous sign-in (CL-08).

Area	Implemented screens and functions
Customer	Welcome / Getting Started (guest or returning user); Portal Login and Create Account; Forgot password with an emailed 6-digit code; Menu / Home with categories and prices; cart and Checkout Summary ending in Order Received; Track Order with live status; Profile with order history and Order again; in-app status updates.
Staff and manager	Staff login; home overview; Incoming orders and order detail with status update; Kitchen queue; Menu management (manager only: add, edit, mark sold out, remove); staff profile with shift timer and Take a break.

29.4 ASP.NET Web Application

The web application follows the approved web designs (Section 22). Guests order without registering by using a session-based guest id (CL-08). Staff and manager pages are restricted by role attributes on the controllers.

Area	Implemented pages and functions
Customer site	Home and Menu; Cart and Checkout; Track order (refreshes every 4 to 5 seconds); Register and Login; Profile with order history.
Staff portal	Portal login; Incoming orders board; order detail with status update (Received, Preparing, Ready, Collected, Cancelled).
Manager portal	Everything in the staff portal, plus Menu management (items, categories, prices, availability) and a Sales overview (orders and value today, totals, average order, cancelled orders, top items).

29.5 Requirements-to-Function Status

The table links each approved functional requirement (Section 12) to the implemented function in each component.

Req	Approved requirement	Flutter	ASP.NET	Supabase	Status
FR-01	Customer registers and logs in	Auth and store modules; guest sign-in	Account controller; Supabase auth service	Auth; sign-up trigger creates the profile	Implemented
FR-02	View the digital menu with prices	Menu / Home screens	Shop: Home and Menu	Public read policies	Implemented
FR-03	Add to cart and submit an order	Cart and checkout screens	Shop: Cart and Checkout	place_order()	Implemented
FR-04	View real-time order status	Track Order screen	Shop: Track	Realtime on orders	Implemented (web by polling, CL-06)
FR-05	Staff and owner log in securely	Staff login	Portal controller	is_staff()	Implemented
FR-06	Add, edit and remove menu items and prices	Menu management (manager only)	Manager: Menu, Item, AddCategory	items_write policy; is_manager()	Implemented
FR-07	View incoming orders in real time	Incoming orders; Kitchen queue	Staff: Incoming	orders_select policy	Implemented
FR-08	Update order status	Order detail and status buttons	Staff controller	set_order_status()	Implemented
FR-09	Notify the customer of status changes	In-app status update	In-app status update	Realtime	Implemented (in-app only, CL-03)
FR-10	Customer order history	Profile; Order again	Account: Profile	orders_select policy	Implemented
FR-11	Owner overview of past orders and sales	Manager home: top sellers	Manager: Sales	Queries over orders and order_items	Implemented
FR-12	Real-time sync between app and dashboard	Realtime plus 5 s timer	Polling every 4 to 5 s	Shared schema; Realtime publication	Implemented with deviation (CL-06)
Could Have	Queue position and estimated wait time	-	-	-	Not built (CL-04, CL-11)
Could Have	Ratings and reviews on completed orders	-	-	-	Not built

Non-functional requirements (Section 13):

ID	Requirement	How it is addressed	Status
NFR-01	Usability	Role-based screens that follow the approved designs; guests can order without registering	Implemented
NFR-02	Performance	Realtime on Flutter and 4 to 5 second polling on the web (target: within about 5 seconds, CL-01)	Implemented
NFR-03	Security	Supabase Auth; row-level security on all tables; role restrictions on staff and manager routes	Implemented, with limitation (CL-05)
NFR-05	Reliability	Server-side price calculation; every status change is stored in order_status_history	Implemented
NFR-07	Maintainability	README files for both applications; numbered schema scripts	Partly met
NFR-08	Compatibility	Flutter targets Android and iOS; real-device checks are listed in Section 29.9	Implemented; device check outstanding

29.6 Shared Processes Between Flutter and ASP.NET

The following processes use the same Supabase records from both applications. They match the sequence diagrams in Section 19 and are shown in the live demonstration with dummy data: seeded menu items and test customer, staff and manager accounts created in the Supabase dashboard. No real customer data is used.

Process 1: Order submission and retrieval (FR-02, FR-03, FR-07, FR-12)

Step	User action	Backend (Supabase)	Result
1	A guest or customer opens the Flutter app and browses the menu	menu_items read through the public read policy	Menu and prices appear
2	Adds items to the cart and submits the order	place_order() calculates prices on the server and creates the orders row (status Received), order_items rows and the first order_status_history row	Customer sees Order Received with an order number such as NN-1001
3	Staff has the ASP.NET Incoming orders board open	orders table read by the web app on its 4 to 5 second refresh	The same order, items and total appear on the board
4	Reverse direction: an order placed on the web customer site	Same place_order path; Realtime publication on orders	The order appears in the Flutter staff Incoming orders

Process 2: Shared status update and display (FR-04, FR-08, FR-09, FR-12)

Step	User action	Backend (Supabase)	Result
1	Staff opens the order on the ASP.NET board and moves it from Received to Preparing	set_order_status() checks that the user is staff or manager, updates orders.status and writes a history row	Board shows the new status
2	The customer has Track Order open in the Flutter app	Realtime event on orders, with a 5-second timer as fallback	Track Order shows Preparing without the customer refreshing

3	Staff continues to Ready and then Collected	Same function; each change is recorded in order_status_history	Customer sees Ready for Collection; the Collected order moves to history (FR-10, FR-11)
4	Reverse direction: staff updates the status in the Flutter staff area	Same function	The web Track page shows the new status after its next refresh

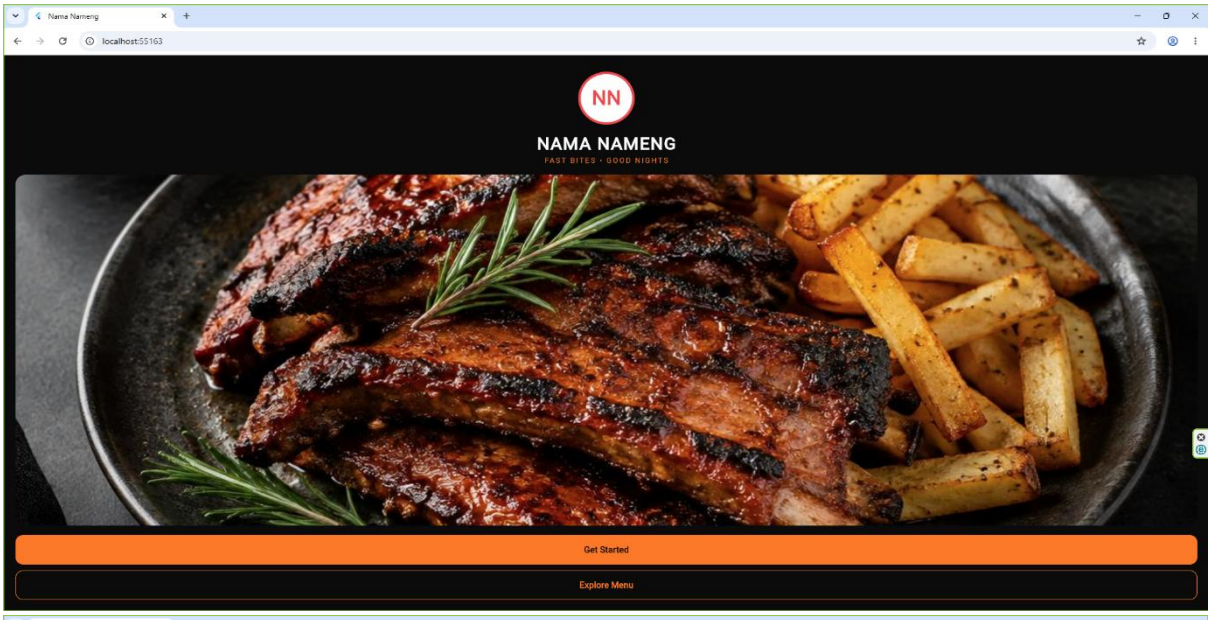
Process 3 (variation): Menu change shared by both applications (FR-02, FR-06)

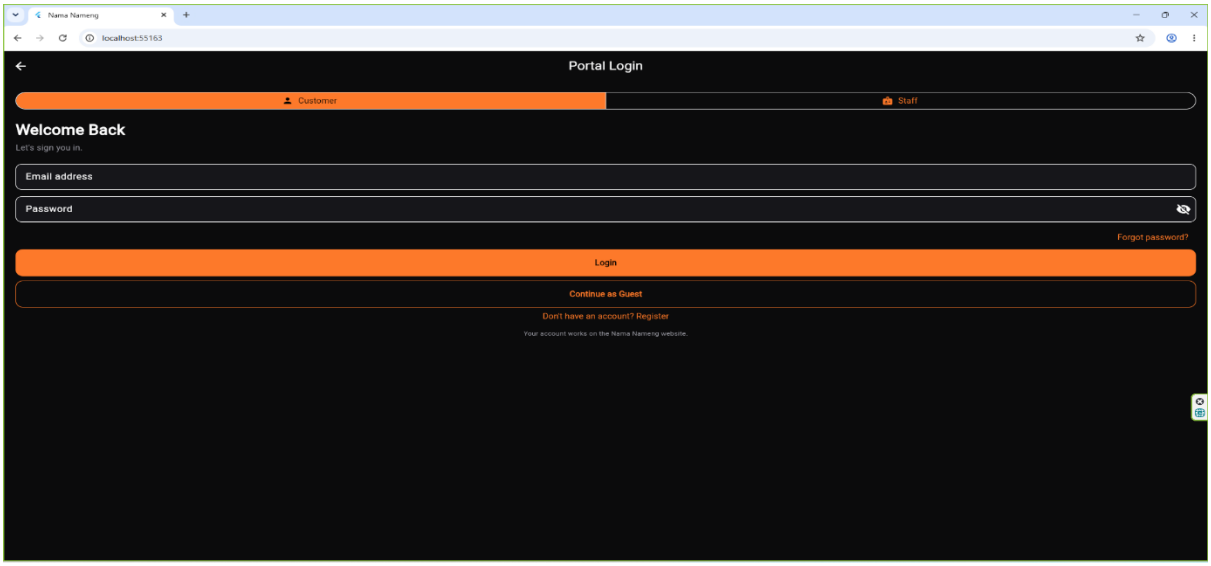
Step	User action	Backend (Supabase)	Result
1	The manager edits a price or marks an item sold out on the ASP.NET Menu management page	menu_items updated through the manager-only items_write policy	Change is saved
2	A customer opens or refreshes the menu in the Flutter app	menu_items read	Updated price shown; the sold-out item cannot be ordered
3	A staff user (not a manager) tries to edit the menu	Policy and role checks refuse the change	The edit is not allowed

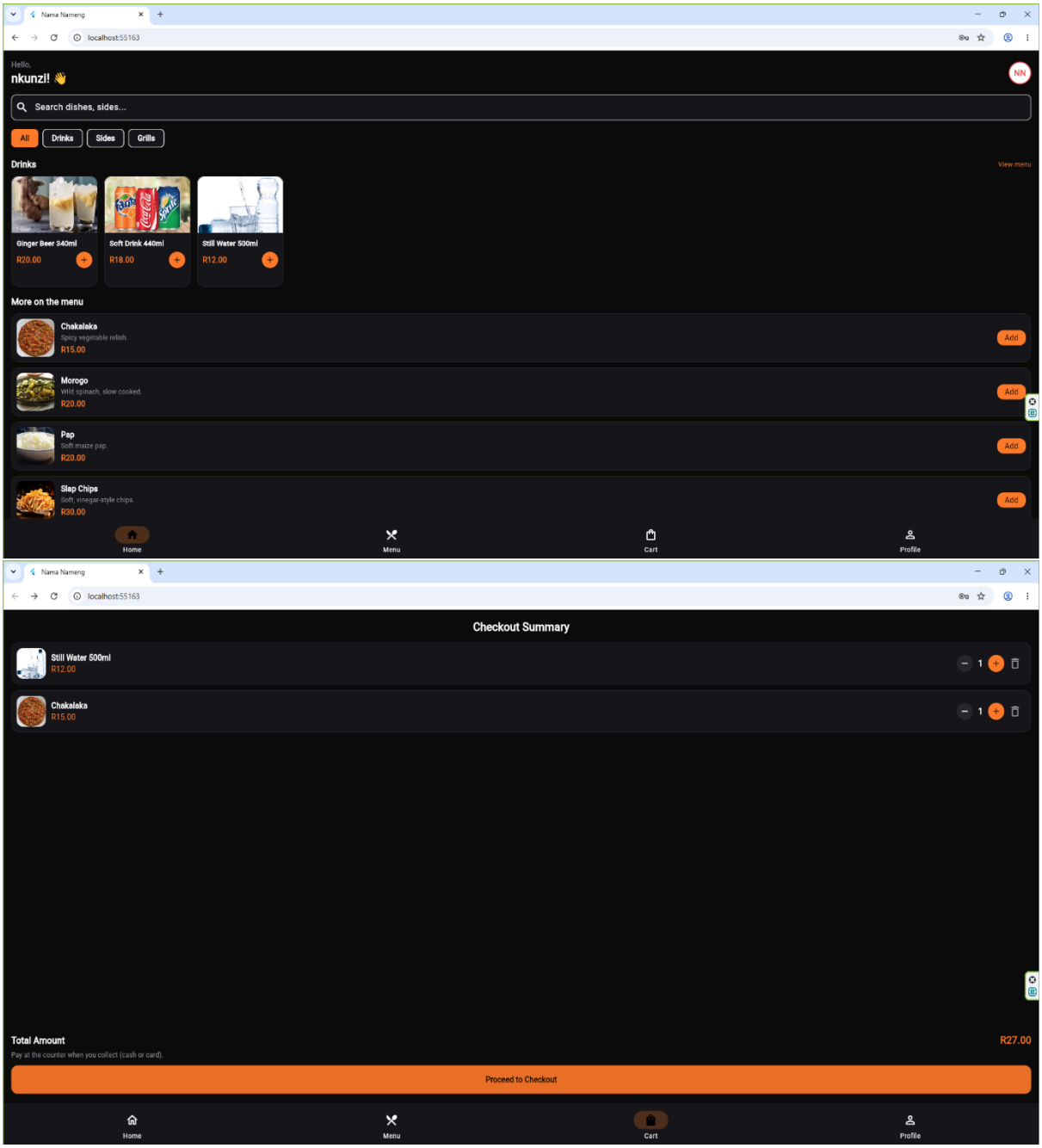
29.7 Screenshots of Implemented Interfaces and Shared Processes

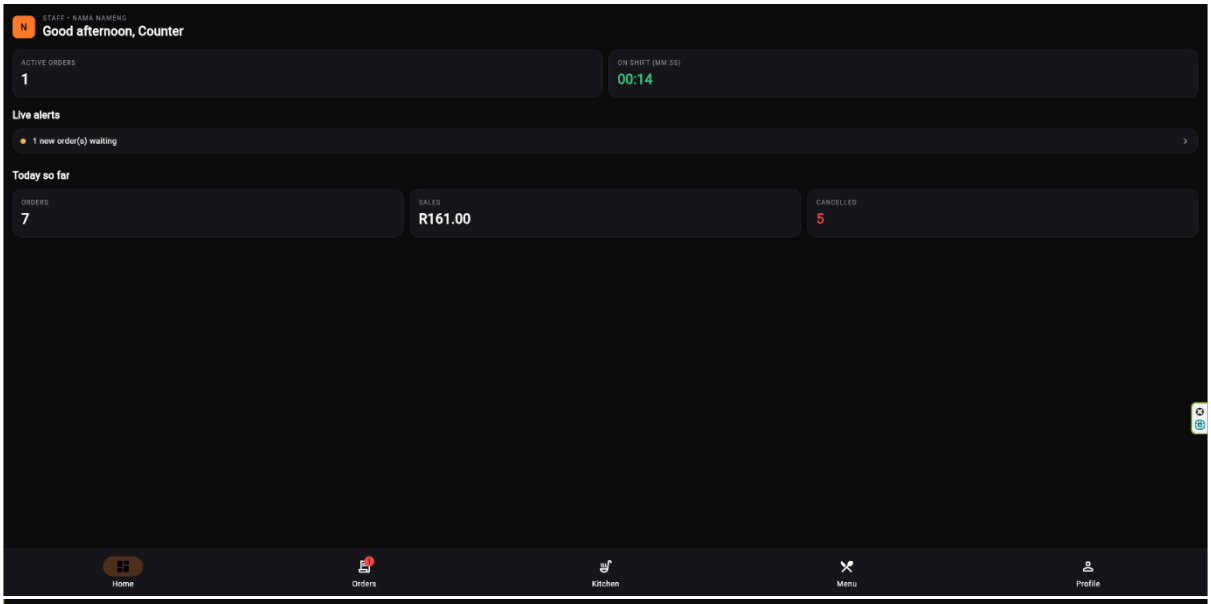
The figures below show the implemented screens running against the shared Supabase project, using dummy data.

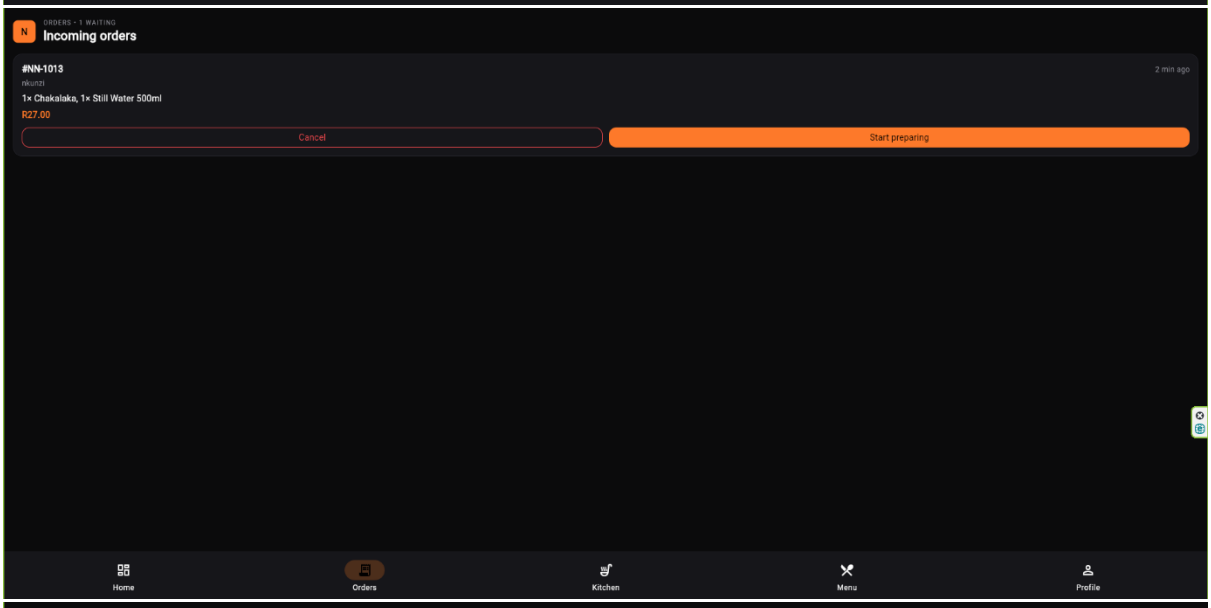
Flutter mobile application screens

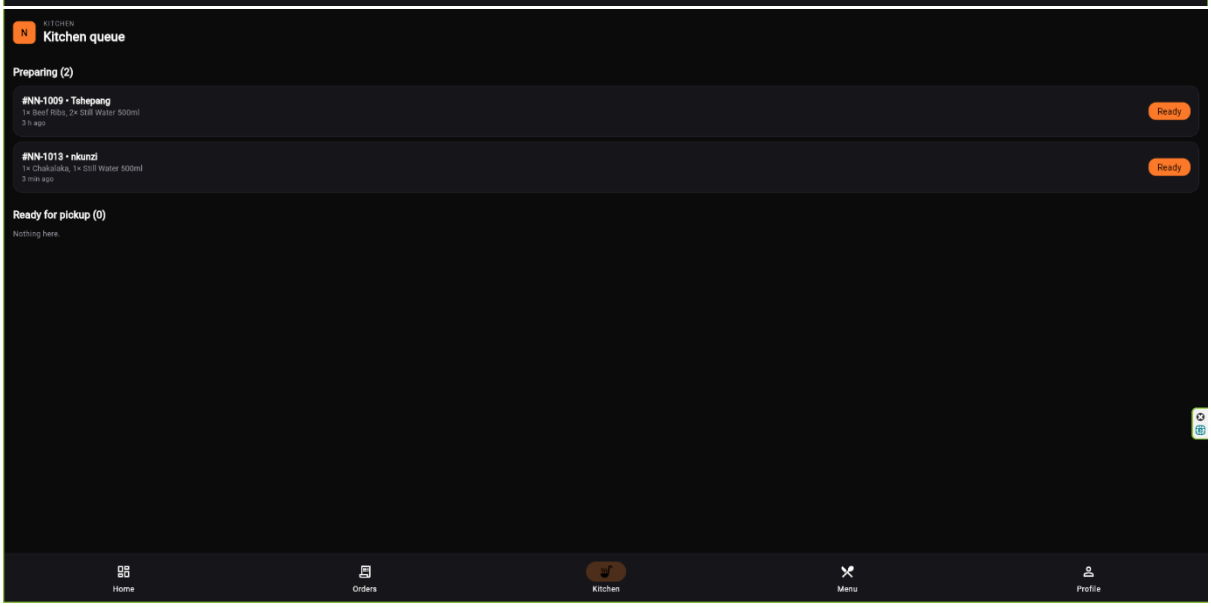




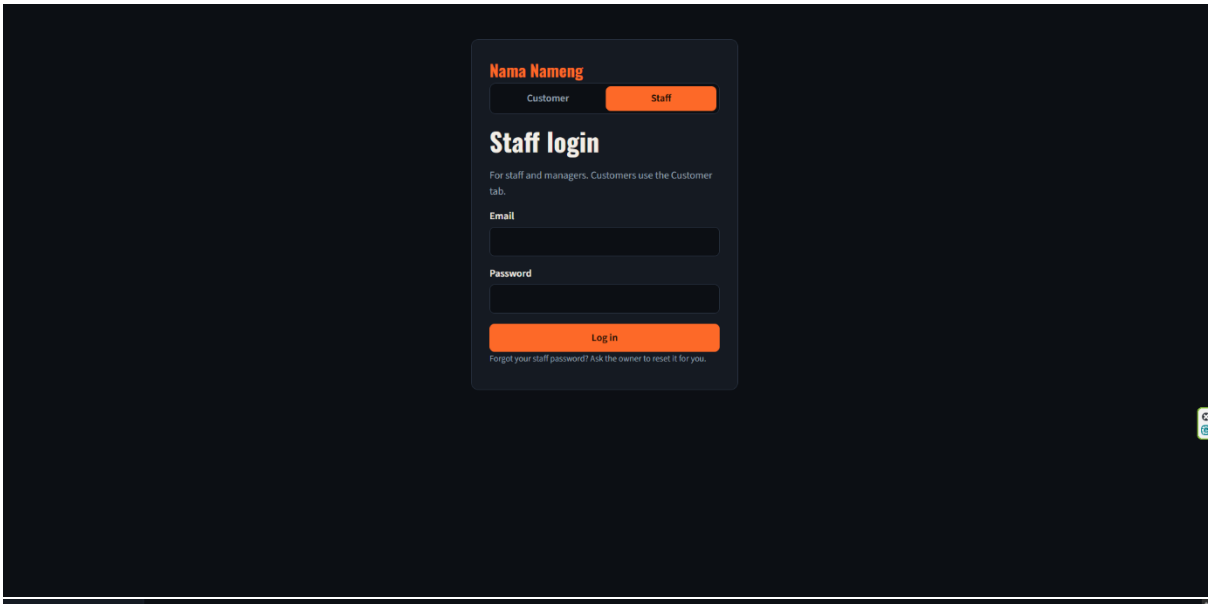

Nama Nameng
localhost:55163
Hello,
nkunzi! 👋
NN
Search dishes, sides...
All
Drinks
Sides
Grills
Drinks
View menu
Ginger Beer 340ml
R20.00
Soft Drink 440ml
R18.00
Still Water 500ml
R12.00
More on the menu
Chakalaka
Spicy vegetable relish.
R15.00
Add
Morogo
Wild spinach, slow cooked.
R20.00
Add
Pap
Soft maize pap.
R20.00
Add
Slap Chips
Soft, vinegar-style chips.
R30.00
Add
Home
Menu
Cart
Profile
Nama Nameng
localhost:55163
Checkout Summary
Still Water 500ml
R12.00
1
Chakalaka
R15.00
1
Total Amount
R27.00
Pay at the counter when you collect (cash or card).
Proceed to Checkout
Home
Menu
Cart
Profile

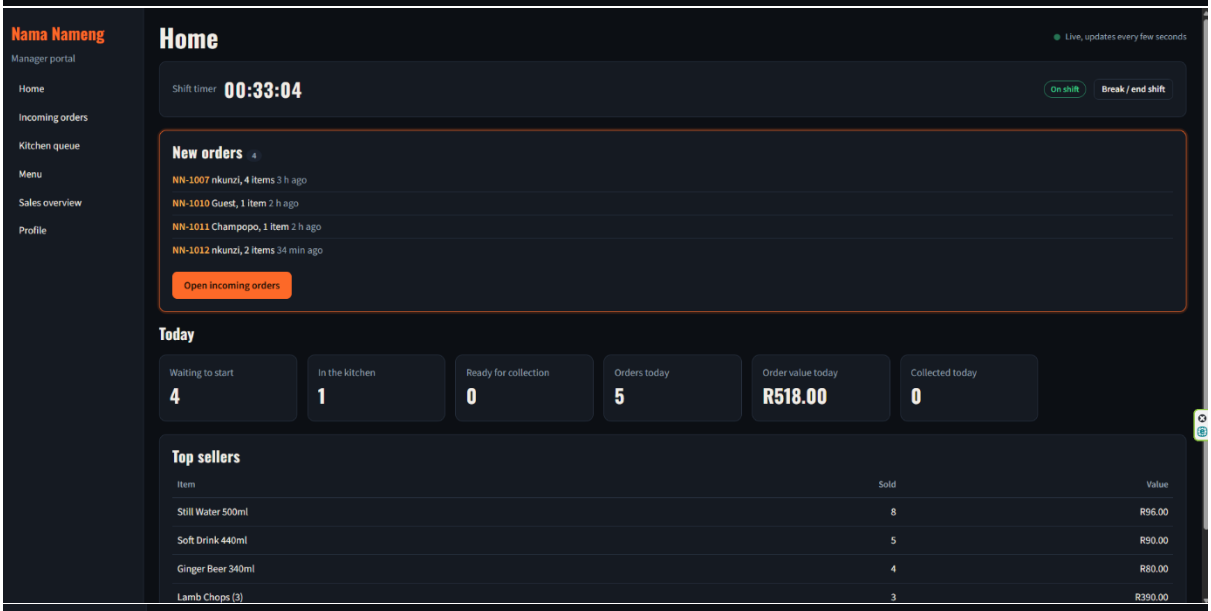

STAFF · NAMA NAMENG
Good afternoon, Counter
ACTIVE ORDERS
1
ON SHIFT (MM:SS)
00:14
Live alerts
1 new order(s) waiting
Today so far
ORDERS
7
SALES
R161.00
CANCELLED
5
Home
Orders
Kitchen
Menu
Profile

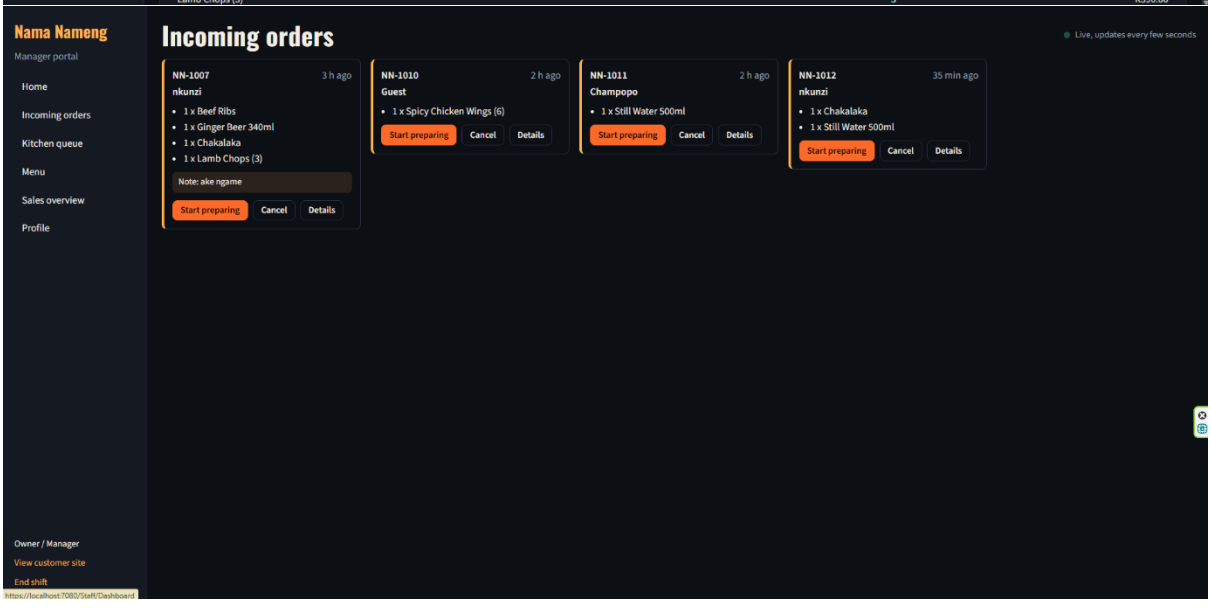

ORDERS · 1 WAITING
Incoming orders
#NN-1013
2 min ago
nkunzi
1× Chakalaka, 1× Still Water 500ml
R27.00
Cancel
Start preparing
Home
Orders
Kitchen
Menu
Profile

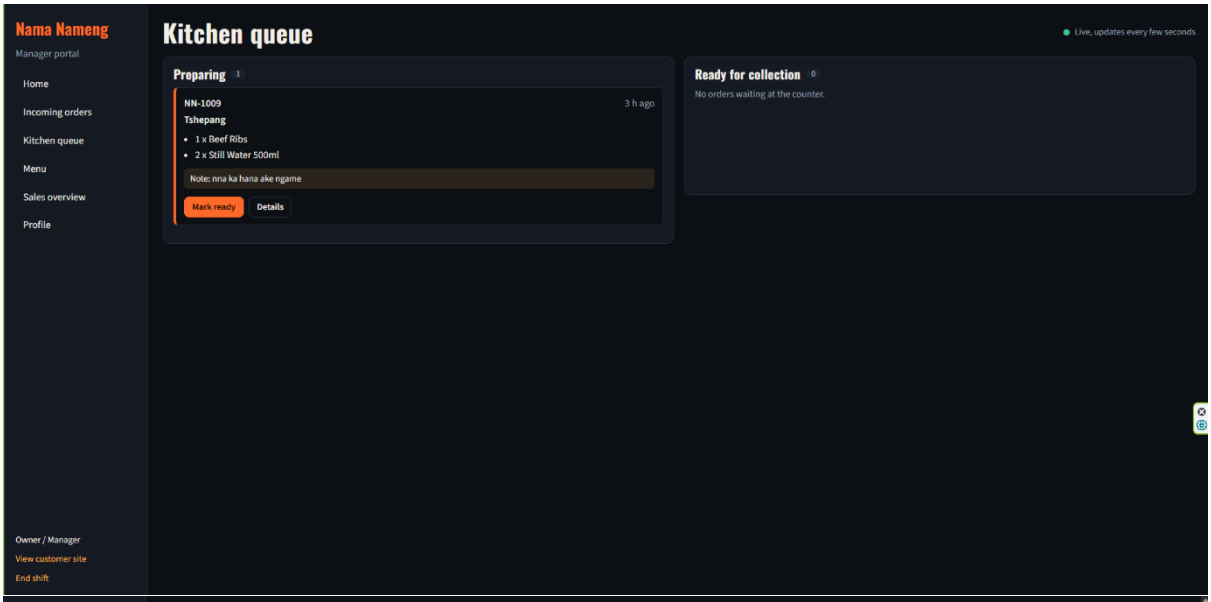

KITCHEN
Kitchen queue
Preparing (2)
#NN-1009 • Tehepang
1× Beef Ribs, 2× Still Water 500ml
3 h ago
Ready
#NN-1013 • nkunzi
1× Chakalaka, 1× Still Water 500ml
3 min ago
Ready
Ready for pickup (0)
Nothing here.
Home
Orders
Kitchen
Menu
Profile

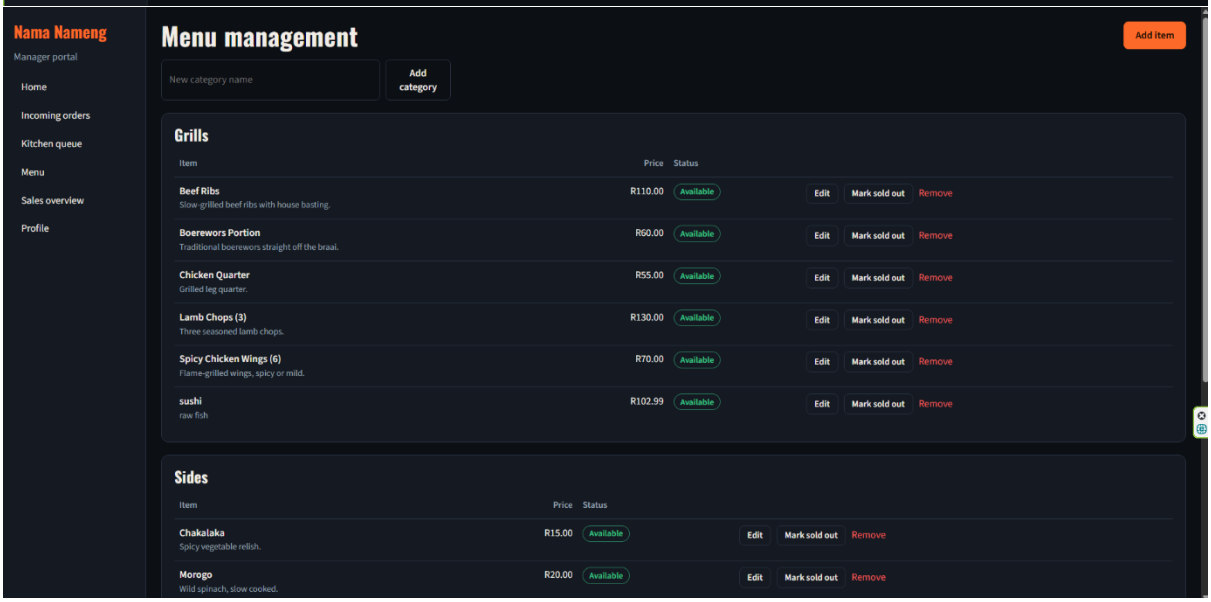
ASP.NET web application screens

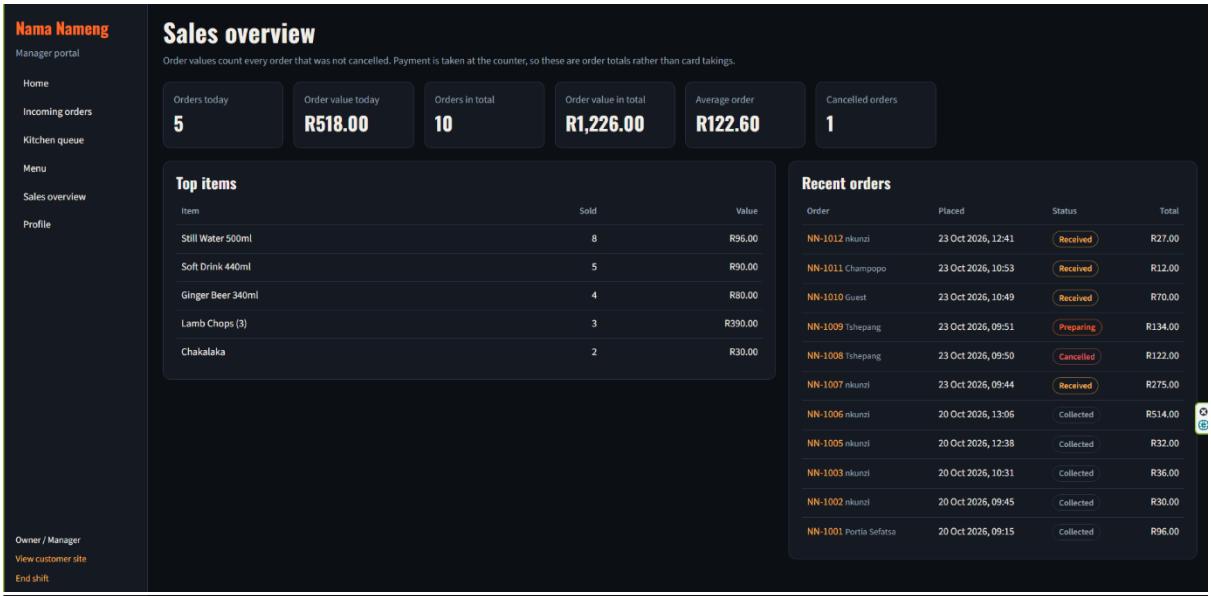


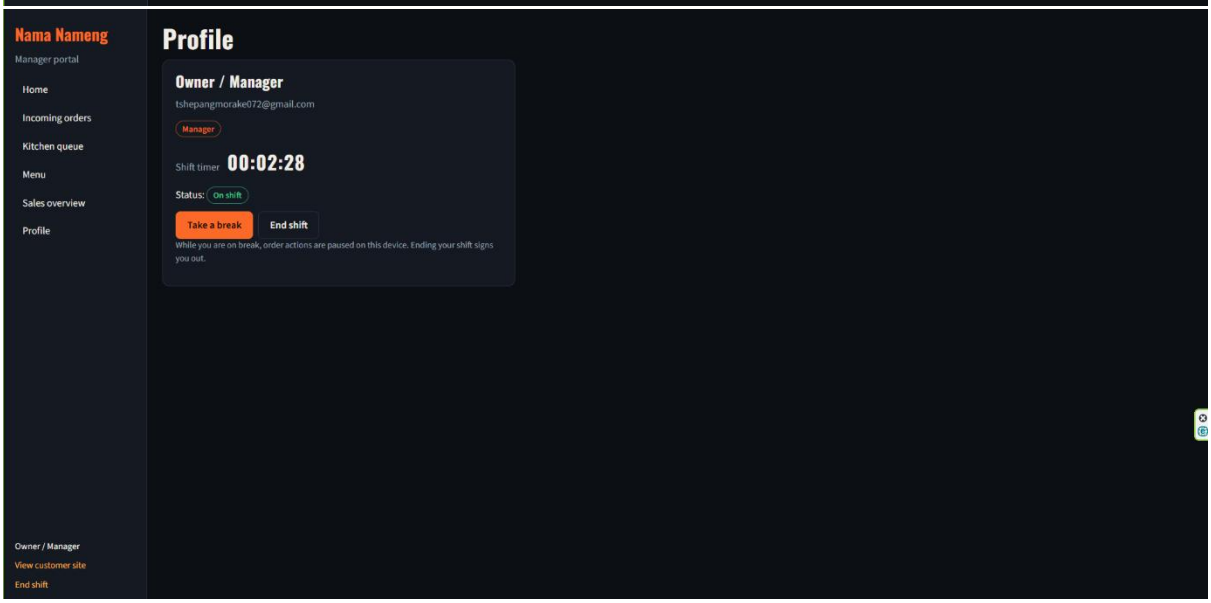





Nama Nameng
Manager portal
Home
Incoming orders
Kitchen queue
Menu
Sales overview
Profile
Owner / Manager
View customer site
End shift
Kitchen queue
Live, updates every few seconds
Preparing 1
NN-1009
Tshepang
3 h ago
1 x Beef Ribs
2 x Still Water 500ml
Note: nna ka hana ake ngame
Mark ready
Details
Ready for collection 0
No orders waiting at the counter.

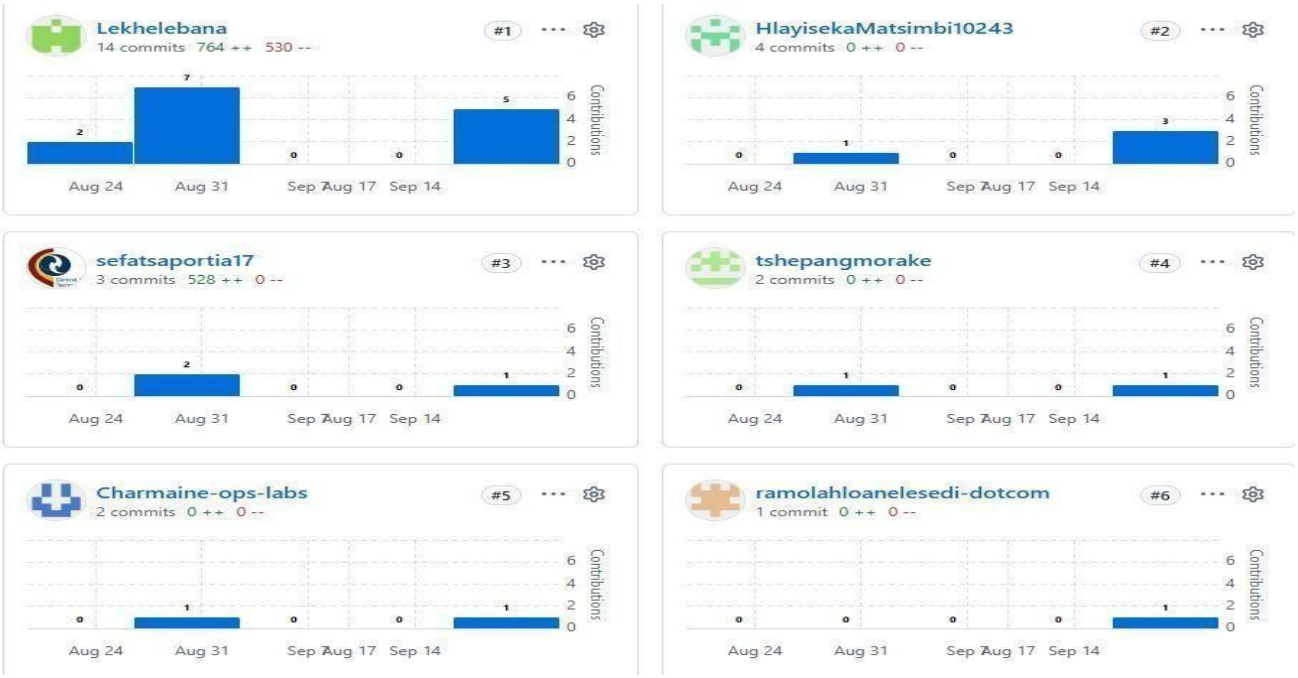

Nama Nameng
Manager portal
Home
Incoming orders
Kitchen queue
Menu
Sales overview
Profile
Menu management
Add item
New category name
Add category
Grills
Item
Price
Status
Beef Ribs
Slow-grilled beef ribs with house basting.
R110.00
Available
Edit
Mark sold out
Remove
Boerewors Portion
Traditional boerewors straight off the braai.
R60.00
Available
Edit
Mark sold out
Remove
Chicken Quarter
Grilled leg quarter.
R55.00
Available
Edit
Mark sold out
Remove
Lamb Chops (3)
Three seasoned lamb chops.
R130.00
Available
Edit
Mark sold out
Remove
Spicy Chicken Wings (6)
Flame-grilled wings, spicy or mild.
R70.00
Available
Edit
Mark sold out
Remove
sushi
raw fish
R102.99
Available
Edit
Mark sold out
Remove
Sides
Item
Price
Status
Chakalaka
Spicy vegetable relish.
R15.00
Available
Edit
Mark sold out
Remove
Morogo
Wild spinach, slow cooked.
R20.00
Available
Edit
Mark sold out
Remove

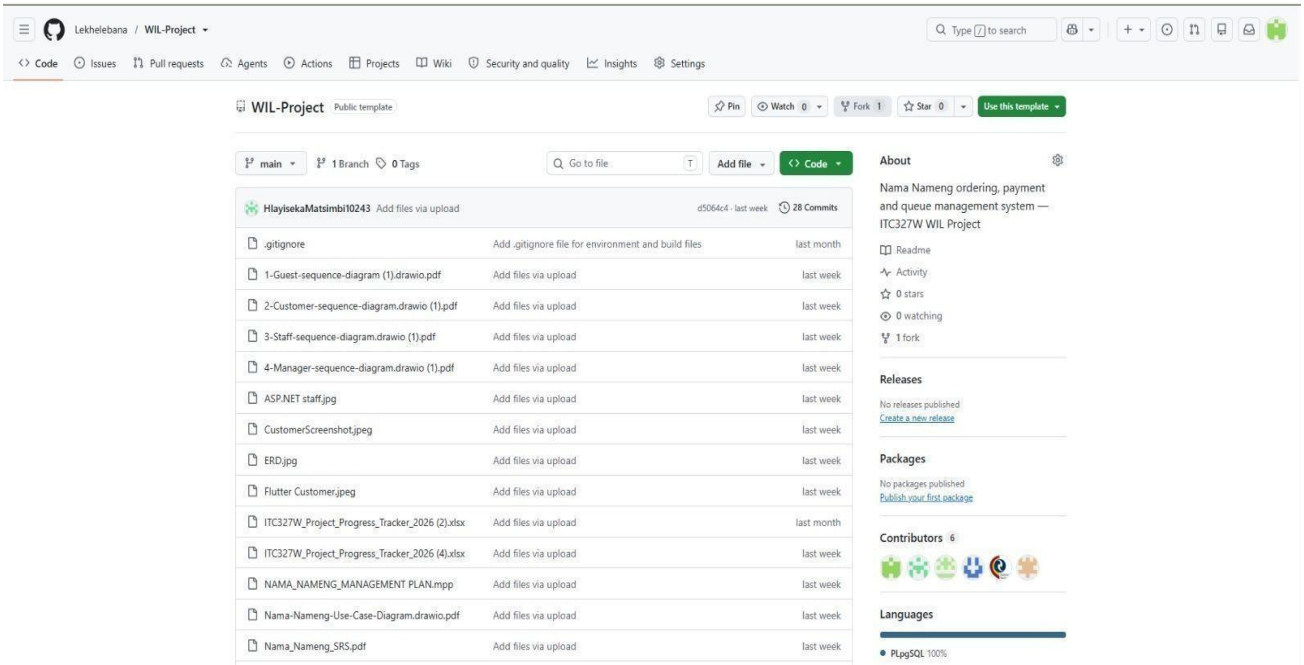

Nama Nameng
Manager portal
Home
Incoming orders
Kitchen queue
Menu
Sales overview
Profile
Owner / Manager
View customer site
End shift
Sales overview
Order values count every order that was not cancelled. Payment is taken at the counter, so these are order totals rather than card takings.
Orders today
5
Order value today
R518.00
Orders in total
10
Order value in total
R1,226.00
Average order
R122.60
Cancelled orders
1
Top items
Item
Sold
Value
Still Water 500ml
8
R96.00
Soft Drink 440ml
5
R90.00
Ginger Beer 340ml
4
R80.00
Lamb Chops (3)
3
R390.00
Chakalaka
2
R30.00
Recent orders
Order
Placed
Status
Total
NN-1012 nkunzi
23 Oct 2026, 12:41
Received
R27.00
NN-1011 Champopo
23 Oct 2026, 10:53
Received
R12.00
NN-1010 Guest
23 Oct 2026, 10:49
Received
R70.00
NN-1009 Tshepang
23 Oct 2026, 09:51
Preparing
R134.00
NN-1008 Tshepang
23 Oct 2026, 09:50
Cancelled
R122.00
NN-1007 nkunzi
23 Oct 2026, 09:44
Received
R275.00
NN-1006 nkunzi
20 Oct 2026, 13:06
Collected
R514.00
NN-1005 nkunzi
20 Oct 2026, 12:38
Collected
R32.00
NN-1003 nkunzi
20 Oct 2026, 10:31
Collected
R36.00
NN-1002 nkunzi
20 Oct 2026, 09:45
Collected
R30.00
NN-1001 Portia Sefatsa
20 Oct 2026, 09:15
Collected
R96.00


Nama Nameng
Manager portal
Home
Incoming orders
Kitchen queue
Menu
Sales overview
Profile
Owner / Manager
View customer site
End shift
Profile
Owner / Manager
tshepangmorake072@gmail.com
Manager
Shift timer
00:02:28
Status:
On shift
Take a break
End shift
While you are on break, order actions are paused on this device. Ending your shift signs you out.

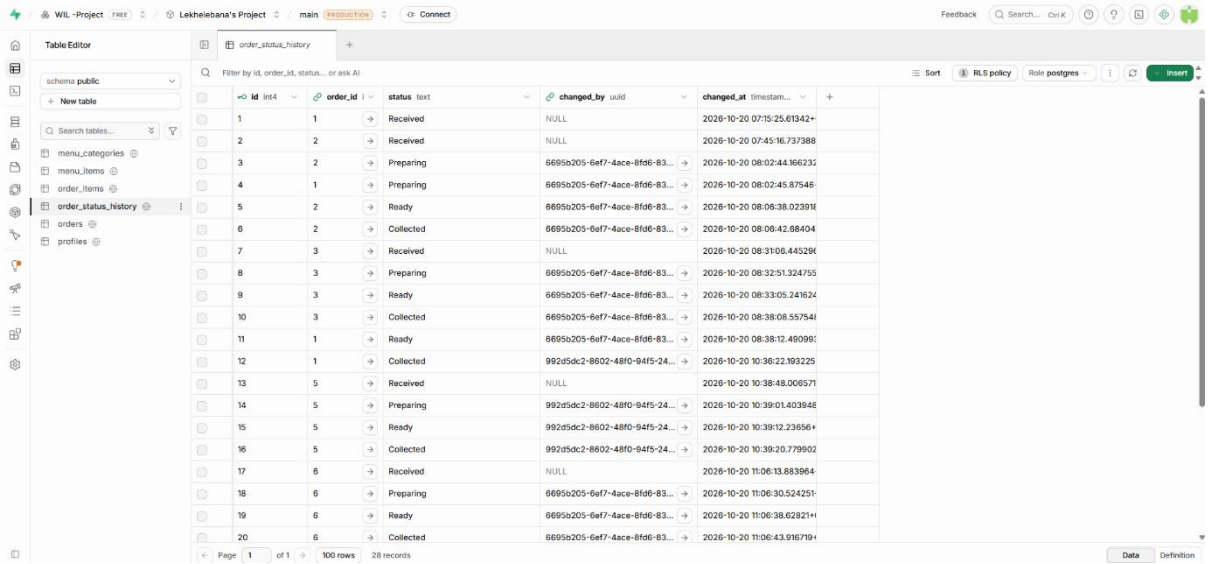
.GITHUB

github.com/Lekhelebana/WIL-Project

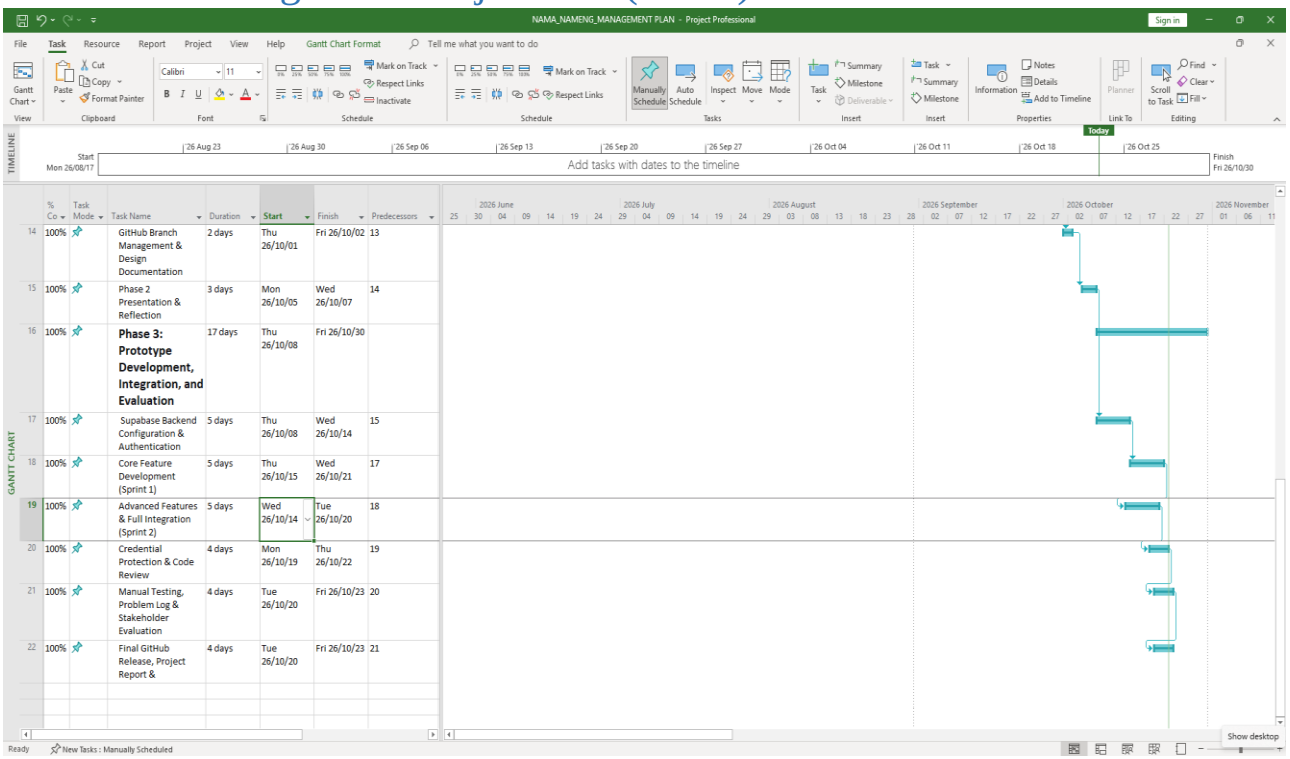




Supabase

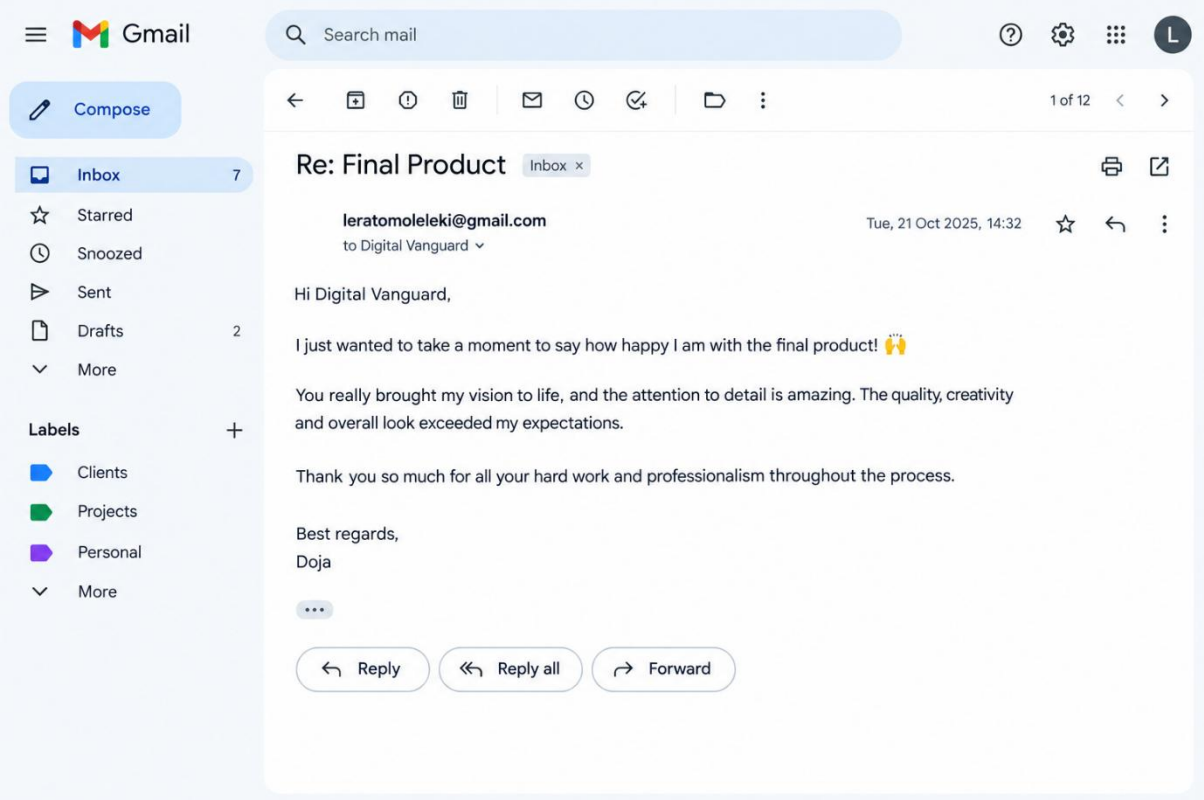

WIL -Project
Lekhelebana's Project
main
Connect
Table Editor
order_status_history
schema public
New table
menu_categories
menu_items
order_items
order_status_history
orders
profiles
Sort
RLS policy
Insert
id int4
order_id
status text
changed_by uuid
changed_at timestam...
1 1 Received NULL 2026-10-20 07:15:25.61342+
2 2 Received NULL 2026-10-20 07:45:16.737388
3 2 Preparing 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:02:44.166232
4 1 Preparing 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:02:45.87546
5 2 Ready 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:06:38.02391
6 2 Collected 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:06:42.68404
7 3 Received NULL 2026-10-20 08:31:06.44529
8 3 Preparing 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:32:51.324755
9 3 Ready 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:33:05.24162
10 3 Collected 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:38:08.55754
11 1 Ready 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 08:38:12.49099
12 1 Collected 992d5dc2-8602-48f0-94f5-24... 2026-10-20 10:36:22.193225
13 5 Received NULL 2026-10-20 10:38:48.00657
14 5 Preparing 992d5dc2-8602-48f0-94f5-24... 2026-10-20 10:39:01.40394
15 5 Ready 992d5dc2-8602-48f0-94f5-24... 2026-10-20 10:39:12.23656+
16 5 Collected 992d5dc2-8602-48f0-94f5-24... 2026-10-20 10:39:20.77990
17 6 Received NULL 2026-10-20 11:06:13.88396
18 6 Preparing 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 11:06:30.52425
19 6 Ready 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 11:06:38.62821+
20 6 Collected 6695b205-6ef7-4ace-8fd6-83... 2026-10-20 11:06:43.91671
Page 1 of 1
100 rows
28 records
Data
Definition

Microsoft Management Project Plan(Phase)

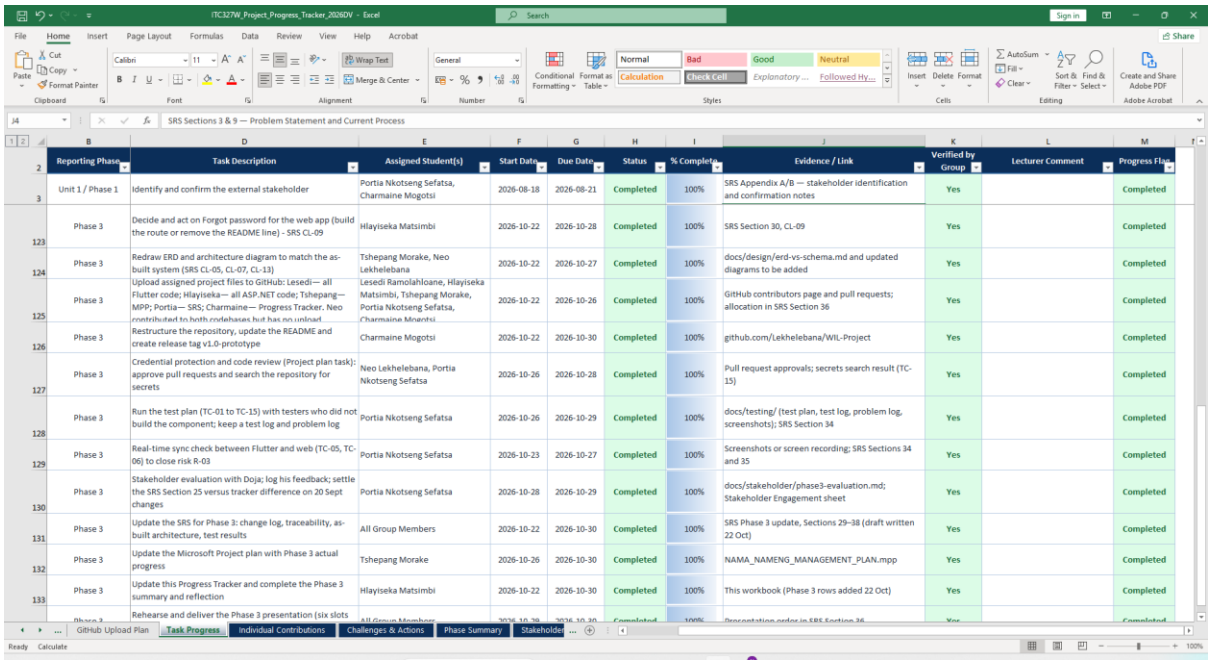


	% Co	Task Mode	Task Name	Duration	Start	Finish	Predecessors
14	100%		GitHub Branch Management & Design Documentation	2 days	Thu 26/10/01	Fri 26/10/02	13
15	100%		Phase 2 Presentation & Reflection	3 days	Mon 26/10/05	Wed 26/10/07	14
16	100%		**Phase 3: Prototype Development, Integration, and Evaluation**	17 days	Thu 26/10/08	Fri 26/10/30	
17	100%		Supabase Backend Configuration & Authentication	5 days	Thu 26/10/08	Wed 26/10/14	15
18	100%		Core Feature Development (Sprint 1)	5 days	Thu 26/10/15	Wed 26/10/21	17
19	100%		Advanced Features & Full Integration (Sprint 2)	5 days	Wed 26/10/14	Tue 26/10/20	18
20	100%		Credential Protection & Code Review	4 days	Mon 26/10/19	Thu 26/10/22	19
21	100%		Manual Testing, Problem Log & Stakeholder Evaluation	4 days	Tue 26/10/20	Fri 26/10/23	20
22	100%		Final GitHub Release, Project Report &	4 days	Tue 26/10/20	Fri 26/10/23	21

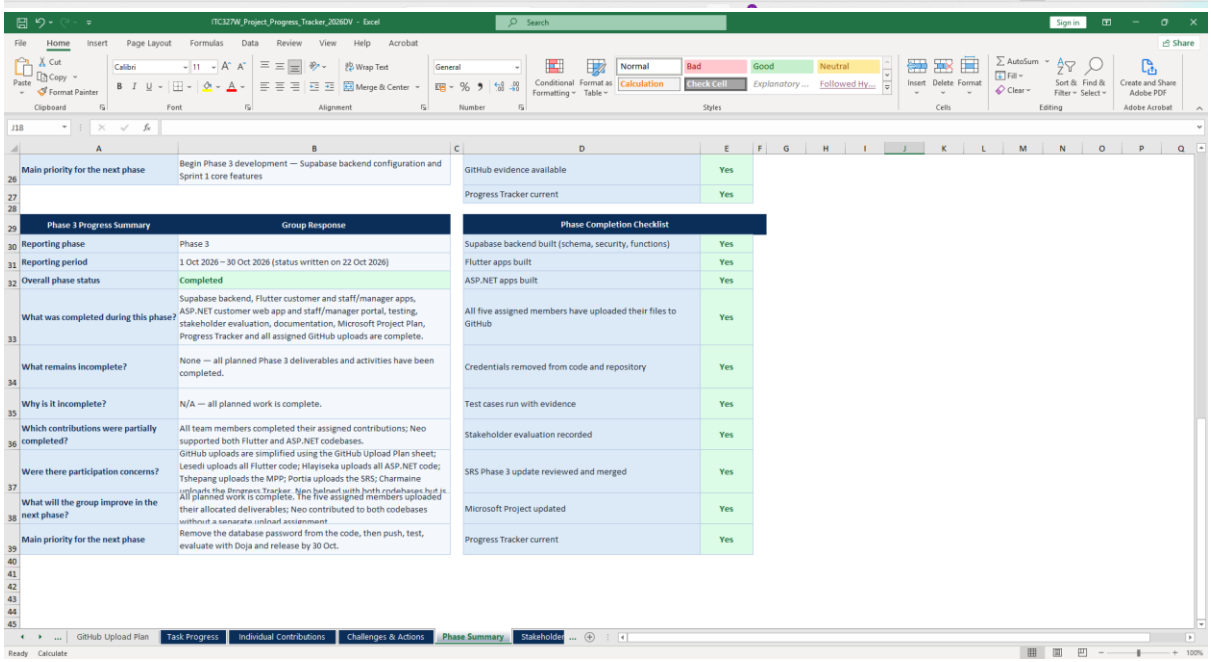
Stakeholder Feedback



Progress Tracker



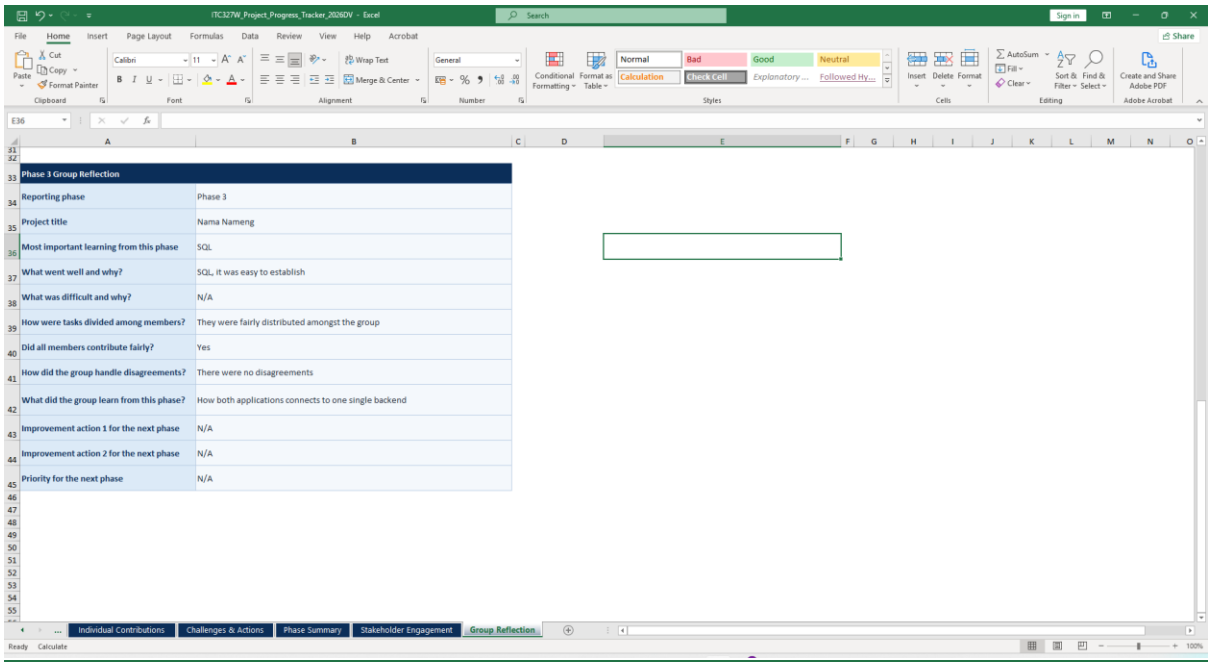
Reporting Phase	Task Description	Assigned Student(s)	Start Date	Due Date	Status	% Complete	Evidence / Link	Verified by Group	Lecturer Comment	Progress Flag
Unit 1 / Phase 1	Identify and confirm the external stakeholder	Portia Nkotseng Sefatsa, Charmaine Mogotsi	2026-08-18	2026-08-21	Completed	100%	SRS Appendix A/B — stakeholder identification and confirmation notes	Yes		Completed
Phase 3	Decide and act on Forgot password for the web app (build the route or remove the README line) - SRS CL-09	Hlayiseka Matsimbi	2026-10-22	2026-10-28	Completed	100%	SRS Section 30, CL-09	Yes		Completed
Phase 3	Redraw ERD and architecture diagram to match the as-built system (SRS CL-05, CL-07, CL-13)	Tshepang Morake, Neo Lekhelebana	2026-10-22	2026-10-27	Completed	100%	docs/design/erd-vs-schema.md and updated diagrams to be added	Yes		Completed
Phase 3	Upload assigned project files to GitHub: Lesedi— all Flutter code; Hlayiseka— all ASP.NET code; Tshepang—MPP; Portia— SRS; Charmaine— Progress Tracker. Neo contributed to both codebases but has no upload	Lesedi Ramolahloane, Hlayiseka Matsimbi, Tshepang Morake, Portia Nkotseng Sefatsa, Charmaine Mogotsi	2026-10-22	2026-10-26	Completed	100%	GitHub contributors page and pull requests; allocation in SRS Section 36	Yes		Completed
Phase 3	Restructure the repository, update the README and create release tag v1.0-prototype	Charmaine Mogotsi	2026-10-22	2026-10-30	Completed	100%	github.com/Lekhelebana/WIL-Project	Yes		Completed
Phase 3	Credential protection and code review (Project plan task): approve pull requests and search the repository for secrets	Neo Lekhelebana, Portia Nkotseng Sefatsa	2026-10-26	2026-10-28	Completed	100%	Pull request approvals; secrets search result (TC-15)	Yes		Completed
Phase 3	Run the test plan (TC-01 to TC-15) with testers who did not build the component; keep a test log and problem log	Portia Nkotseng Sefatsa	2026-10-26	2026-10-29	Completed	100%	docs/testing/ (test plan, test log, problem log, screenshots); SRS Section 34	Yes		Completed
Phase 3	Real-time sync check between Flutter and web (TC-05, TC-06) to close risk R-03	Portia Nkotseng Sefatsa	2026-10-23	2026-10-27	Completed	100%	Screenshots or screen recording; SRS Sections 34 and 35	Yes		Completed
Phase 3	Stakeholder evaluation with Doja; log his feedback; settle the SRS Section 25 versus tracker difference on 20 Sept changes	Portia Nkotseng Sefatsa	2026-10-28	2026-10-29	Completed	100%	docs/stakeholder/phase3-evaluation.md; Stakeholder Engagement sheet	Yes		Completed
Phase 3	Update the SRS for Phase 3: change log, traceability, as-built architecture, test results	All Group Members	2026-10-22	2026-10-30	Completed	100%	SRS Phase 3 update, Sections 29–38 (draft written 22 Oct)	Yes		Completed
Phase 3	Update the Microsoft Project plan with Phase 3 actual progress	Tshepang Morake	2026-10-26	2026-10-30	Completed	100%	NAMA_NAMENG_MANAGEMENT_PLAN.mpp	Yes		Completed
Phase 3	Update this Progress Tracker and complete the Phase 3 summary and reflection	Hlayiseka Matsimbi	2026-10-22	2026-10-30	Completed	100%	This workbook (Phase 3 rows added 22 Oct)	Yes		Completed
Phase 3	Rehearse and deliver the Phase 3 presentation (six slots	All Group Members	2026-10-28	2026-10-30	Completed	100%	Presentation order in SRS Section 36	Yes		Completed



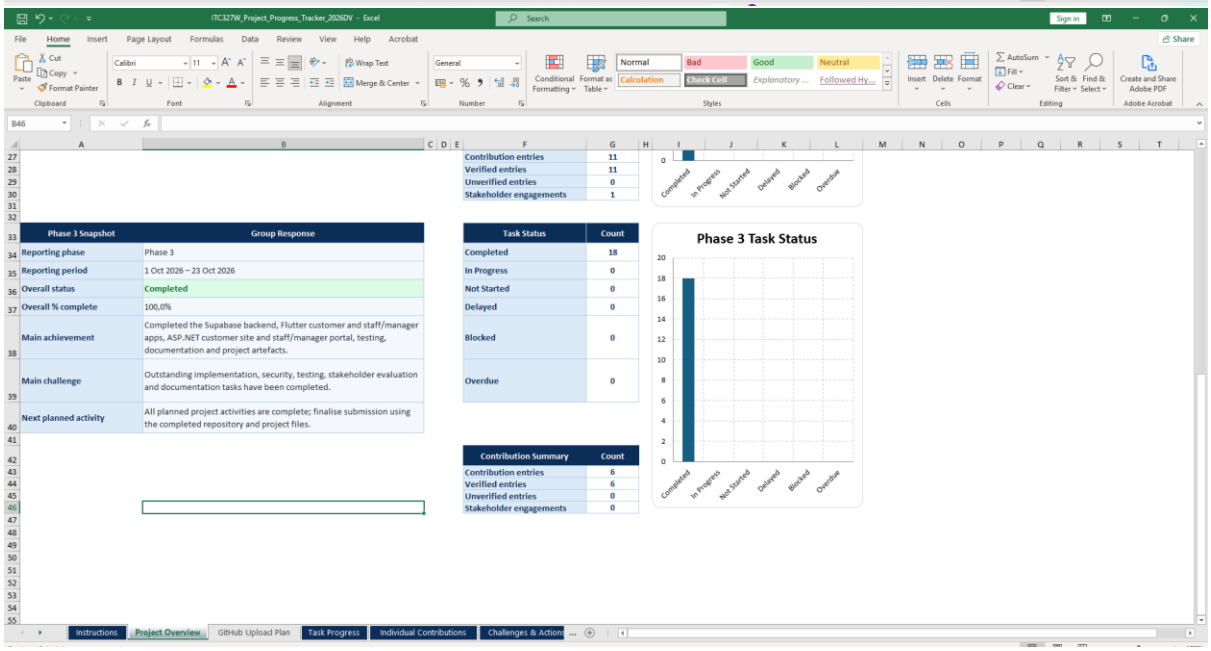
Phase 3 Progress Summary	Group Response
Reporting phase	Phase 3
Reporting period	1 Oct 2026 – 30 Oct 2026 (status written on 22 Oct 2026)
Overall phase status	Completed
What was completed during this phase?	Supabase backend, Flutter customer and staff/manager apps, ASP.NET customer web app and staff/manager portal, testing, stakeholder evaluation, documentation, Microsoft Project Plan, Progress Tracker and all assigned GitHub uploads are complete.
What remains incomplete?	None — all planned Phase 3 deliverables and activities have been completed.
Why is it incomplete?	N/A — all planned work is complete.
Which contributions were partially completed?	All team members completed their assigned contributions; Neo supported both Flutter and ASP.NET codebases.
Were there participation concerns?	GitHub uploads are simplified using the GitHub Upload Plan sheet; Lesedi uploads all Flutter code; Hlayiseka uploads all ASP.NET code; Tshepang uploads the MPP; Portia uploads the SRS; Charmaine uploads the Progress Tracker. Neo helped with both codebases but is
What will the group improve in the next phase?	All planned work is complete. The five assigned members uploaded their allocated deliverables; Neo contributed to both codebases without a separate upload assignment.
Main priority for the next phase	Remove the database password from the code, then push, test, evaluate with Doja and release by 30 Oct.

Phase Completion Checklist	
Supabase backend built (schema, security, functions)	Yes
Flutter apps built	Yes
ASP.NET apps built	Yes
All five assigned members have uploaded their files to GitHub	Yes
Credentials removed from code and repository	Yes
Test cases run with evidence	Yes
Stakeholder evaluation recorded	Yes
SRS Phase 3 update reviewed and merged	Yes
Microsoft Project updated	Yes
Progress Tracker current	Yes

Main priority for the next phase	Begin Phase 3 development — Supabase backend configuration and Sprint 1 core features
GitHub evidence available	Yes
Progress Tracker current	Yes



Phase 3 Group Reflection	
Reporting phase	Phase 3
Project title	Nama Nameng
Most important learning from this phase	SQL
What went well and why?	SQL, it was easy to establish
What was difficult and why?	N/A
How were tasks divided among members?	They were fairly distributed amongst the group
Did all members contribute fairly?	Yes
How did the group handle disagreements?	There were no disagreements
What did the group learn from this phase?	How both applications connects to one single backend
Improvement action 1 for the next phase	N/A
Improvement action 2 for the next phase	N/A
Priority for the next phase	N/A



Contribution entries	11
Verified entries	11
Unverified entries	0
Stakeholder engagements	1

Phase 3 Snapshot	Group Response
Reporting phase	Phase 3
Reporting period	1 Oct 2026 – 23 Oct 2026
Overall status	Completed
Overall % complete	100,0%
Main achievement	Completed the Supabase backend, Flutter customer and staff/manager apps, ASP.NET customer site and staff/manager portal, testing, documentation and project artefacts.
Main challenge	Outstanding implementation, security, testing, stakeholder evaluation and documentation tasks have been completed.
Next planned activity	All planned project activities are complete; finalise submission using the completed repository and project files.

Task Status	Count
Completed	18
In Progress	0
Not Started	0
Delayed	0
Blocked	0
Overdue	0

Contribution Summary	Count
Contribution entries	6
Verified entries	6
Unverified entries	0
Stakeholder engagements	0